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Improving surface quality and profile accuracy of five-axis machining via vector error coupling modeling and compensation

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Abstract

In five-axis machining, ensuring geometric accuracy while maximizing efficiency is a critical challenge. Conventional adaptive interpolation strategies often rely on a single local curvature threshold, which overlooks the complex interaction between Tool Center Point (TCP) path errors and Tool Axis Vector (TAV) orientation errors, leading to inefficient use of computationally expensive interpolation modes. To address this, this paper proposes a novel adaptive hybrid interpolation strategy based on a vector error coupling model. The strategy uses a "dual-check" decision logic: first, local curvature is measured to identify low-risk regions, where efficient linear interpolation (LERP) is applied. In high-curvature regions, a subsequent check predicts the total geometric error by summing the chord height and swing errors, capturing the additive or subtractive coupling effects linked to surface concavity or convexity. The system switches to high-precision Non-Uniform Rational B-Splines (NURBS) and spherical linear interpolation (SLERP) only if the coupling is additive, or if the predicted error in a subtractive mode exceeds the tolerance. This logic enables efficient LERP even in high-curvature regions as long as the total error remains within limits. Physical machining experiments on 6061 aluminum alloy validate the strategy, demonstrating a 91.7% reduction in maximum measured profile error (from 42.31 μm to 3.52 μm) and a 54% improvement in surface roughness. Concurrently, this strategy reduces computation time by nearly 40% compared to a full NURBS and SLERP approach, enhancing efficiency without compromising precision. This research offers a practical solution that significantly improves both surface integrity and processing efficiency in high-performance machining.

Keywords: Five-axis machining, Vector error coupling, Chord height error, Machining accuracy, Adaptive hybrid interpolation

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1 Introduction

Five-axis computer numerical control (CNC) machining is a pivotal technology for addressing the challenges in manufacturing complex-surface components [1–3], such as aero-engine blades and precision molds [4,5]. Through the multi-degree-of-freedom motion, it enables efficient, single-setup machining of intricate structures [6,7]. However, a core technical difficulty remains: maximizing machining efficiency while ensuring exceptionally high trajectory accuracy. As the bridge between Computer-Aided Design and Computer-Aided Manufacturing (CAD/CAM) design blueprints and physical motion, the toolpath interpolation algorithm plays a decisive role in determining final machining quality and time costs. Recent advancements in the field highlight the stringent requirements for surface integrity, particularly for difficult-to-cut materials. For instance, Cai et al. [8,9] and Chen et al. [10,11] investigated advanced processing strategies for titanium alloys and ceramics, demonstrating that optimizing surface morphology is critical for functional performance. Similarly, Yan et al. [12] emphasized the importance of surface modification in reducing friction and wear. Against this background of pursuing extreme surface quality, optimizing the interpolation algorithm becomes a fundamental prerequisite.

Traditional interpolation strategies typically employ a single method, often leading to a trade-off between efficiency and accuracy. On one hand, LERP is computationally simple and fast, but it produces significant profile error (i.e., chord error) when processing paths with high curvature by approximating curves with straight lines [13,14]. On the other hand, high-order interpolation algorithms such as Non-Uniform Rational B-Splines (NURBS) can precisely fit complex curves and

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ensure trajectory smoothness, but their real-time interpolation process faces unique challenges. The complex nonlinear relationship between a NURBS curve's parameter and its actual arc length makes it difficult to precisely control the feed rate while satisfying geometric constraints like chord error and centripetal acceleration [15,16]. Applying such a computationally intensive method in geometrically simple, flat regions would unnecessarily sacrifice machining efficiency [17,18].

To resolve this conflict between precision and efficiency, a substantial body of research has been conducted. Addressing the challenges of NURBS interpolation, many studies have focused on optimizing look-ahead feedrate scheduling algorithms to dynamically adjust velocity and satisfy multiple constraints during machining. The core of these algorithms involves preprocessing the toolpath to generate a smooth and efficient velocity profile based on geometric features (e.g., curvature) and machine dynamics limitations (e.g., maximum velocity, acceleration, and jerk of each axis). Common methods include using S-shaped acceleration/deceleration (ACC/DEC) curves for smooth velocity transitions [19,20] and incorporating bidirectional scanning techniques to plan and adjust cornering speeds to avoid constraint violations [21]. Some research has explored more advanced strategies, such as directly planning the velocity profile using B-spline curves [22], establishing analytical models to ensure strict adherence to drive axis kinematic constraints [23], and using methods like morphological filtering to handle velocity profile fluctuations [24]. Concurrently, another stream of research has focused on enhancing the precision of LERP. Some scholars posit that accurate contour error estimation is a prerequisite for effective compensation. For example, Wang et al. [13] proposed an accelerated iteration algorithm to rapidly solve for the contour error point, while Peng et al. [25] utilized Dynamic Time Warping (DTW) to establish a more accurate mapping between command and actual trajectories for complex paths. Kunzi et al. [26] and Man et al. [27] explored contour error evaluation from the perspectives of minimum zone fitting and direct visual measurement, respectively. Building on error estimation, scholars have proposed various compensation control strategies. Xiao et al. [28] constructed a model reference adaptive control for pre-compensation of predicted contour errors. In contrast, Wei et al. [29] and Wen et al. [14] developed regional and online compensation methods, respectively, to directly correct the toolpath through real-time error detection. Furthermore, some studies have focused on proactively avoiding errors at their source. Tang et al. [7] investigated a vision-based geometric error self-calibration method to improve the machine tool's fundamental accuracy. In parallel, researchers have proposed various methods for identifying and decoupling geometric errors introduced by rotary axes in five-axis machining. For instance, Guo et al. [30] used spatial trajectory planning for error decoupling and identification, Lyu et al. [31] utilized an R-test device for measurement, and Song et al. [32] proposed a vision-based identification scheme. Meanwhile, Tang et al. [33] employed filters to smooth corners, thereby reducing contour error while satisfying kinematic constraints. To further enhance stability, recent studies by Hu et al. [34,35] introduced self-synchronization and self-balancing theories to suppress unfavorable vibrations. Recent studies have further pushed the boundaries of high-precision interpolation and real-time processing. For instance, Xu et al. [36] proposed an optimized dual NURBS interpolation strategy that utilizes a master-slave curve mechanism to synchronize tool position and orientation, significantly reducing chord errors in complex surface machining. In parallel, Qiu et al. [37] developed an efficient online look-ahead algorithm for geometric simulation, demonstrating the critical importance of computational efficiency and real-time feasibility in intelligent CNC systems. However, notwithstanding these kinematic and dynamic advancements, a fundamental limitation persists: most existing methods focus on optimizing error control within a single interpolation mode. This leads to a critical processing bottleneck: to guarantee surface integrity and avoid tool marks, machines are often forced to run at an overly conservative (slow) feed rate, even in regions where it may not be necessary. This inefficiency stems from an underexplored phenomenon: the complex vectorial interaction between different error sources—specifically, the TCP chord height error and the TAV swing error—remains largely underexplored. More critically, the fundamental relationship between this error coupling (i.e., whether the errors are additive or subtractive) and the local surface concavity or convexity has not been systematically investigated. Consequently, existing adaptive mechanisms remain limited by a rigid reliance on single-curvature thresholds. They operate on the conservative assumption that high curvature universally demands high-precision interpolation, overlooking the potential for error cancellation in convex regions (subtractive coupling). This leads to the unnecessary activation of computationally expensive interpolation modes in these 'safe' zones, creating a significant efficiency gap that current methods fail to bridge.

To address the aforementioned technical bottleneck, this paper proposes a novel "dual-check" adaptive hybrid interpolation strategy based on a vector error coupling model. The core concept of this method is to move beyond a single curvature threshold by analyzing the vectorial interaction between the TCP chord height error and the TAV swing error to make a more intelligent decision. This mechanism allows the system to safely use high-efficiency interpolation even in high-curvature zones, maximizing machining efficiency while guaranteeing precision. This paper validates this new strategy

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through extensive physical machining experiments on 6061 aluminum alloy, demonstrating significant, measurable improvements in both final surface quality and overall processing time.

The strategy's logic first employs local curvature as a primary filter: in low-curvature regions, both TCP and TAV utilize efficient LERP interpolation. Upon entering a high-curvature region, the system triggers a second check by analyzing the error coupling mode. In high-risk "additive coupling" regions (typically concave surfaces), the system switches to the high-precision NURBS+SLERP combination. Critically, however, in "subtractive coupling" regions (typically convex surfaces), the strategy intelligently maintains LERP+LERP interpolation as long as the predicted total error remains within tolerance. This mechanism allows the system to safely use high-efficiency interpolation even in high-curvature zones, maximizing machining efficiency while guaranteeing precision.

The remainder of this paper is organized as follows: Section 2 introduces the kinematic model. Section 3 elaborates on the traditional error sources. Section 4 details the proposed adaptive hybrid interpolation strategy. Section 5 validates and compares the performance of the method through a series of simulations and physical machining experiments on 6061 aluminum alloy. Finally, Section 6 discusses the measured improvements in machining accuracy and surface roughness, and Section 7 concludes the paper and discusses potential directions for future research.

2 Machine Tool Kinematic Model

2.1 Structure and Coordinate System of the A-C Dual-turntable Five-axis CNC Machine

The subject of this study is the A-C dual-turntable five-axis CNC machine, whose structure and kinematic coordinate system are shown in Figure 1. The machine consists of three translational axes (*X, Y, Z*) and two rotational axes (*A, C*), allowing the tool to perform complex five degrees of freedom motion relative to the workpiece. A key feature of this machine is that the workpiece moves along the two rotational axes, while the tool only undergoes translation along the three translational axes. This configuration provides good rigidity and is well-suited for the precise machining of complex parts.

2.2 Machine Tool Kinematic Forward and Inverse Solution

In the kinematic analysis of the A-C dual-turntable five-axis CNC Machine, the transformation between the machine coordinate system (MCS) and the workpiece coordinate system (WCS) is fundamental, as shown in Figure 1(b). Homogeneous coordinate transformations, rotation matrices, and translation matrices are used to describe the movement of the TCP and the TAV.

First, in the workpiece coordinate system, the homogeneous coordinates of the tool's center point and TAV at a certain moment can be represented as:

$$(x, y, z, i, j, k),$$

where x, y, z are the position coordinates of the tool center, and i, j, k are the direction cosines of the TAV.

By transforming from the workpiece coordinate system to the machine coordinate system, the tool's position coordinates and orientation in the machine coordinate system are obtained:

$$(X, Y, Z, A, C),$$

where X, Y, Z are the tool's position in the machine coordinate system, and A, C represent the rotation angles around the X and Z axes, respectively.

To simplify the calculation, it is assumed that the rotation axes of the machine intersect, and the workpiece coordinate system and the C-axis coordinate system overlap. In the A-C dual-turntable machine model, using homogeneous coordinate transformations, the mutual mapping relationship between the MCS and WCS can be derived. The forward kinematic transformation from MCS to WCS is given by Eq. (1) [38]:

$$\begin{cases} x = X\cos\theta_C + Y\sin\theta_A\sin\theta_C + Z\cos\theta_A\cos\theta_C, \\ y = X\sin\theta_C - Y\cos\theta_A\cos\theta_C + Z\sin\theta_A\cos\theta_C, \\ z = Y\sin\theta_A + Z\cos\theta_A, \end{cases} \quad (1)$$

where θ_A and θ_C are the rotation angles of the machine along the X and Z axes, respectively, and X, Y, Z are the position coordinates in the machine coordinate system.

Based on the machine's rotation and translation, the tool axis unit vector i, j, k in the workpiece coordinate system can be expressed as:

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$$\begin{cases} i = \sin\theta_A \sin\theta_C, \\ j = -\sin\theta_A \cos\theta_C, \\ k = \cos\theta_A. \end{cases} \tag{2}$$

These unit vectors describe the rotation direction of the tool. Using Eqs. (1), (2), the transformation between the MCS and WCS can be accomplished.

For inverse kinematics, the goal is to calculate the machine's joint angles (rotation axis angles) and the position in the machine coordinate system, based on the desired tool position and orientation in the workpiece coordinate system (TCP and TAV). This process involves calculating the rotation angles θ_A and θ_C, which depend on the values of i, j, and k:

$$\theta_A = \arccos(k),$$
$$\theta_C = \begin{cases} \text{Arbitrary value} & \text{when } k = \pm 1, \\ \dfrac{\pi}{2}, & \text{when } k \neq \pm 1, j = 0, i > 0, \\ -\dfrac{\pi}{2}, & \text{when } k \neq \pm 1, j = 0, i < 0, \\ \arctan\left(\dfrac{-i}{j}\right), & \text{otherwise.} \end{cases} \tag{3}$$

Eq. (4) describes the conversion of the tool center point coordinates from the workpiece coordinate system to the machine coordinate system:

$$\begin{cases} X = x\cos\theta_C + y\sin\theta_C, \\ Y = -x\cos\theta_A \sin\theta_C + y\cos\theta_A \cos\theta_C + z\sin\theta_A, \\ Z = z\cos\theta_A - x\sin\theta_A. \end{cases} \tag{4}$$

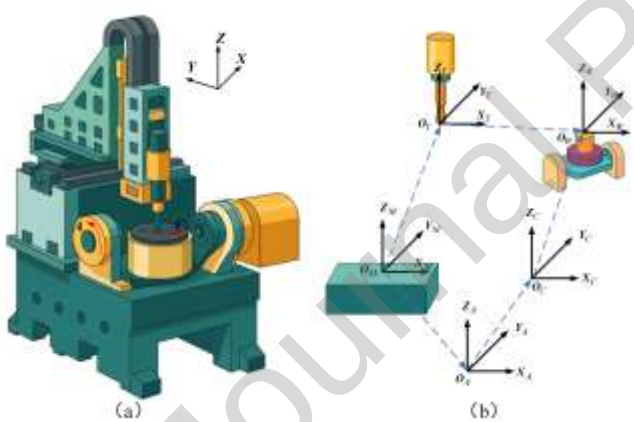

Figure 1 Kinematic analysis and structural design of A-C dual-turntable five-axis CNC machine: (a) Structural schematic of A-C dual-turntable five-axis CNC machine, (b) Forward and inverse kinematic diagram of A-C dual-turntable five-axis CNC machine

Since the curvature-adaptive algorithm operates in the workpiece coordinate system (WCS), both the forward and inverse kinematic transformations are essential for converting tool positions and orientations between the workpiece coordinate system (WCS) and the machine coordinate system (MCS). The forward transformation converts data from the WCS to the MCS, while the inverse transformation performs the reverse, converting data from the MCS to the WCS.

This dual transformation process allows the algorithm to be effectively applied to the machine tool, ensuring that the required motion in the MCS is achieved, even when the initial calculations are performed in the WCS.

Whether the data is obtained from the machine coordinate system or the workpiece coordinate system—such as from machine feedback or simulation software—the algorithm can accommodate both. If the data is in the MCS, it is first converted to the WCS using the inverse transformation. Then, the algorithm is applied to generate the required motion, followed by a conversion back to the MCS using the forward transformation. If the data is already in the WCS, the algorithm can be directly applied, followed by conversion to the MCS.

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3 Traditional Error Analysis in Five-axis Machining

3.1 Chord Height Error Modeling for TCP Path

To achieve adaptive selection of tool trajectory interpolation methods, it is necessary to accurately assess the local geometric characteristics of the trajectory. This paper uses the "three-point method" to perform geometric analysis of the tool trajectory and employs the chord height error (δ) as a key metric for measuring interpolation accuracy. Figure 2 illustrates the geometric relationships between consecutive nodes on the 3D spatial trajectory, where A_{i-1}, A_i, and A_{i+1} represent three consecutive nodes on the trajectory.

For the three non-collinear points A_{i-1}, A_i, and A_{i+1} in space, they uniquely determine an external circle. The radius of this external circle, R, defines the curvature radius of the trajectory in the region. The curvature κ is defined as the reciprocal of the curvature radius:

$$\kappa = \frac{1}{R}, \tag{5}$$

where, R can be calculated through the triangle formed by points A_{i-1}, A_i, and A_{i+1} (as shown in Figure 2). The sides a, b, and c are the sides of the triangle, and S is the area of the triangle. The circumradius R can be calculated as:

$$R = \frac{abc}{4S}, \tag{6}$$

where a, b, and c are the sides of the triangle, and S is the area of the triangle. The area S can be calculated using Heron's equation, as follows:

$$\begin{cases} s = \dfrac{a+b+c}{2} (semi\text{-}perimeter), \\ S = \sqrt{s(s-a)(s-b)(s-c)}. \end{cases} \tag{7}$$

Eq. (7) calculates the area of the triangle based on its three sides, which can then be used to calculate the curvature of the circumradius. By substituting Eqs. (5), (6), and (7), the curvature can be expressed as:

$$\kappa = \frac{\sqrt{(a+b+c)(-a+b+c)(a-b+c)(a+b-c)}}{abc}. \tag{8}$$

The chord height error δ is the maximum vertical distance between the arc and its corresponding chord. To calculate the chord height error, we consider the geometry formed by the arc segment between nodes A_{i-1} and A_i, with the center of the circle at O_i and the chord length as a, and the chord height error as δ, as shown in the enlarged view in Figure 2.

In this geometric relationship, the chord height error δ satisfies the Pythagorean theorem with the curvature radius R and the chord length a:

$$R^2 = \left(\frac{a}{2}\right)^2 + (R-\delta)^2. \tag{9}$$

By solving this Pythagorean theorem formula, we can obtain the precise calculation formula for the chord height error δ:

$$\delta = R - \sqrt{R^2 - \left(\frac{a}{2}\right)^2}. \tag{10}$$

By substituting Eq. (5) into Eq. (10), the equation can be transformed into:

$$\delta = \frac{1}{\kappa} - \sqrt{\left(\frac{1}{\kappa}\right)^2 - \left(\frac{a}{2}\right)^2}. \tag{11}$$

This establishes the traditional relationship between path curvature and chord height error, which has conventionally served as a primary metric for assessing TCP interpolation accuracy.

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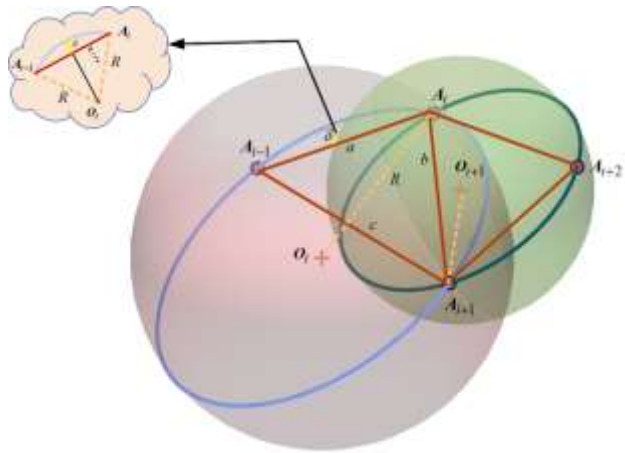

Figure 2 Geometric relationship and chord height error calculation in the local region of a trajectory

3.2 Swing Error Modeling for Tool Orientation

Following the analysis of the tool path contour error in Section 3.1, this section will focus on another critical error source in five-axis machining: the swing error caused by tool orientation changes. Inappropriate orientation interpolation algorithms can severely impact the smoothness of the machining process and the final surface quality.

To visually analyze this error, this paper contrasts two core orientation interpolation algorithms: the ideal SLERP and the simplified LERP.

3.2.1 Swing Error Modeling for Tool Orientation

To clearly demonstrate the differences in the kinematic characteristics of the two algorithms, the subsequent analysis and figures in this paper are based on a typical 90-degree orientation change, as shown in Figure 3(a). The starting orientation is defined by the unit vector v_0= [1,0,0], and the ending orientation is defined by the unit vector v_1= [0,1,0].

The total angle of rotation, Ω, between two orientation vectors can be calculated using the dot product. According to the dot product definition, $v_0 \cdot v_1 = \|v_0\|\|v_1\|\cos(\Omega)$. Since orientation vectors are unit vectors, this simplifies to $\cos(\Omega) = v_0 \cdot v_1$. Therefore, the total angle can be found using the arccosine function:

$$\Omega = \arccos(v_0 \cdot v_1). \tag{12}$$

For the example used in this section, Ω = arccos(0) =90°.

3.2.2 Comparative Analysis of Kinematic Characteristics

Figure 3(b) illustrates the comparison of the angular progression between LERP and SLERP for the 90-degree rotation. We can interpret the instantaneous angular velocity by analyzing the slope of the curves.

SLERP (red dashed line): Its angular progression is a straight line, indicating that its slope is constant. Therefore, its angular velocity is constant. This represents an ideal, smooth, constant-velocity motion.

LERP (orange solid line): Its angular progression is a curved line. Near the start (t=0) and end (t=1) points, the curve's slope is gentle, indicating a slower angular velocity. Near the midpoint (t=0.5), the slope is at its steepest, where the angular velocity reaches its maximum. This curve accurately reflects the "slow-fast-slow" kinematic profile of LERP interpolation. This variable velocity is the root cause of its non-smoothness.

3.3 Comprehensive Swing Error Analysis

3.3.1 Pose error

Pose error is a static, geometric error that refers to the maximum spatial deviation between the LERP path (a chord) and the ideal SLERP path (a great-circle arc). This deviation can be quantified by a dimensionless swing error indicator:

$$\varepsilon = \left(1 - \cos\left(\frac{\Omega}{2}\right)\right). \tag{13}$$

This theoretical error is amplified by the lever effect of the tool length L, resulting in a measurable physical swing error, $\boldsymbol{E}_{\mathbf{swing}}$, which can be estimated as:

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$$\boldsymbol{E}_{swing} \approx L \cdot \varepsilon = L \cdot \left(1 - \cos\left(\frac{\Omega}{2}\right)\right). \tag{14}$$

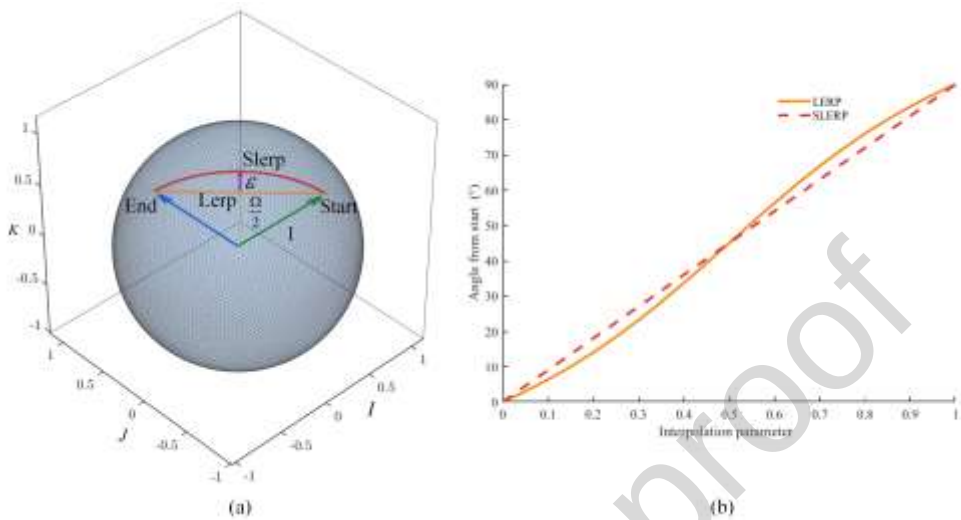

Figure 3 Comparison of LERP and SLERP: (a) Geometric comparison of LERP and SLERP paths, illustrating the Pose Error (ε), (b) Comparison of angular progression and angle smoothness error between LERP and SLERP

3.3.2 Angle smoothness error

Angle smoothness error is a dynamic, kinematic error caused by the variable velocity characteristic of LERP. A kinematic shows that the angular acceleration (α) of SLERP is always zero. In contrast, the angular acceleration of LERP is continuously non-zero, and its discontinuities produce significant Jerk, which is the direct cause of machine vibrations and tool marks on the workpiece.

3.3.3 The Unified Error Model and Adaptive Strategy

The preceding analysis traces both chord height error and swing error back to a single geometric source: the local curvature κ of the toolpath. The relationship between the orientation change and path curvature can be approximated as:

$$\Omega \approx \kappa \cdot \Delta s. \tag{15}$$

By substituting this into Eq. (13), we establish a direct link between path curvature and the swing error indicator:

$$\varepsilon \approx 1 - \cos\left(\frac{\kappa \cdot \Delta s}{2}\right). \tag{16}$$

An important corollary of this error model is that as the path curvature $\kappa \to 0$, the rotation angle $\Omega \to 0$. Using a Taylor series expansion, $\varepsilon \approx \Omega^2/8$. This shows that the swing error risk is proportional to the square of the path curvature. For paths approaching a straight line, the error produced by LERP becomes negligible.

4 A Refined Adaptive Hybrid Interpolation Strategy Based on Error Coupling Analysis

4.1 Foundational Strategy: Curvature-driven Adaptive Interpolation for TCP

To achieve both high efficiency and high precision in five-axis machining, this paper proposes an adaptive hybrid interpolation strategy based on trajectory geometric features. The strategy decouples the tool movement into TCP path motion and TAV attitude motion, and designs independent adaptive interpolation schemes for each. The core of the strategy is to use the local curvature (κ) of the tool path as a unified geometric criterion, enabling intelligent switching between simple LERP and high-precision interpolation (NURBS/SLERP). This approach maximizes feedrate efficiency while

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ensuring machining quality.

4.1.1 Mathematical model of interpolation methods

LERP is the most basic and efficient method for connecting two points. For a given starting point P_0 and endpoint P_1, the parametric expression is:

$$P(u) = (1-u)P_0 + uP_1, u \in [0,1], \tag{17}$$

where, u is the normalized parameter. The computational complexity of LERP is $O(1)$, but when approximating a curved path with a straight line segment, it introduces bow height error (δ).

NURBS is an industrial standard capable of precisely representing complex curves. Its core application in advanced CNC systems lies in the ability to perform dynamic feed rate scheduling based on its strict mathematical definition. This allows the system to reduce speed in areas with large bow height errors in order to minimize the error.

A NURBS curve is defined by the degree p, control points $\{P_i\}$, weights $\{w_i\}$, and the knot vector u:

$$C(u) = \frac{\sum_{i=0}^{n} w_i P_i N_{i,p}(u)}{\sum_{i=0}^{n} w_i N_{i,p}(u)}, \tag{18}$$

where, $N_{i,p}(u)$ is the B-spline basis function, defined by the Cox-de Boor recurrence equation:

$$\begin{cases} N_{i,0}(u) = \begin{cases} 1, & if\ \ u_i \le u < u_{i+1}, \\ 0, & \text{otherwise}, \end{cases} \\ N_{i,p}(u) = \dfrac{u-u_i}{u_{i+p}-u_i} N_{i,p-1}(u) + \dfrac{u_{i+p+1}-u}{u_{i+p+1}-u_{i+1}} N_{i+1,p-1}(u). \end{cases} \tag{19}$$

To perform speed planning, the precise curvature κ(u) at any point on the curve must be obtained, which requires the first derivative $C'(u)$ and the second derivative $C''(u)$ of the curve. Once the derivative vectors are obtained, the curvature is calculated using the first and second derivatives of the curve.

For NURBS curves, the derivatives can be computed by differentiating the summation expression $\sum_{i=0}^{n} w_i P_i N_{i,p}(u)$ and $w(u) = \sum_{i=0}^{n} w_i N_{i,p}(u)$. The expression for $A(u)$ is given by [39]:

$$\begin{cases} C'(u) = \dfrac{A'(u)w(u) - A(u)w'(u)}{[w(u)]^2}, \\ C''(u) = \dfrac{A''(u) - 2C'(u)w'(u) - C(u)w''(u)}{w(u)}, \end{cases} \tag{20}$$

where, $A'(u)$, $A''(u)$, $A'''(u)$, etc., are calculated by differentiating the B-spline basis functions.

After obtaining the first and second derivatives of the curve, the curvature $\kappa(u)$ at any point u on the curve can be calculated using the standard equation:

$$\kappa(u) = \frac{\| C'(u) \times C''(u) \|}{\| C'(u) \|^3}. \tag{21}$$

In Eq. (11), the relationship between the bow height error δ and the curvature κ has already been established. This formula is commonly used to calculate the bow height error in LERP of curves. However, for more complex curves like NURBS, which require advanced calculations, a simplified approach is needed for practical speed planning.

The chord length $a(u)$ corresponding to the interpolation cycle T can be approximated by the feed rate $V(u)$:

$$a(u) \approx V(u) \cdot T. \tag{22}$$

To estimate the bow height error more accurately for NURBS curves, the following approximation is used:

$$\delta(u) \approx \frac{\kappa(u) \cdot V(u)^2 \cdot T^2}{8}. \tag{23}$$

Eq. (23) provides a practical way to calculate the error, considering curvature and feed rate. The maximum allowable

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feedrate $V_{\max}(u)$ is then computed as:

$$V_{\max}(u) \le \frac{\sqrt{8 \cdot \delta_{tol}}}{T} \cdot \frac{1}{\sqrt{\kappa(u)}}. \tag{24}$$

The Look-ahead function helps predict future machine states, generating the optimal feed rate and ensuring the path is executed smoothly and efficiently.

4.1.2 Adaptive hybrid interpolation strategy

Based on the above analysis, this paper proposes an adaptive hybrid interpolation strategy for TCP. The strategy dynamically balances computational efficiency and interpolation accuracy, using the real-time calculated local curvature κ of the trajectory as a criterion. The principle of the strategy is shown in Figure 4.

As shown in Figure 4(a), in the low curvature regions of the trajectory (κ<threshold), the bow height error introduced by the computationally efficient LERP is minimal and can be ignored. However, when the trajectory enters high curvature regions as shown in Figure 4(b) (κ>threshold), LERP introduces significant bow height deviations (δ_1, δ_2). At this point, the interpolation must seamlessly switch to the high-precision NURBS interpolation to ensure machining tolerances are maintained. The resulting hybrid trajectory, shown in Figure 4(c), consists of smoothly connected segments of LERP straight lines and NURBS curves. This adaptive method maximizes machining speed by applying LERP in geometrically simple regions and ensures profile accuracy by using NURBS in more geometrically complex regions. Ultimately, this approach optimizes overall machining efficiency while maintaining the required machining quality. The curvature-driven strategy described above provides a robust and effective baseline for balancing efficiency and precision. By switching to NURBS in high-curvature regions, it successfully controls the primary source of chord height error. However, this model operates under the assumption that high curvature always necessitates high-precision interpolation. Our further analysis reveals that this is not always the case. The final geometric accuracy is determined by the complex interaction of both path and orientation errors. The following sections will introduce a more refined error model that accounts for this interaction, leading to a more intelligent and optimized final strategy.

4.2 Adaptive Hybrid Interpolation Scheme for TAV

Smoothing the TCP path is only half of ensuring machining quality; the smooth transition of the TAV is equally crucial. This section will detail the adaptive interpolation scheme for TAV.

4.2.1 SLERP Algorithm Principle

In order to achieve high efficiency and smooth movement of the TAV, and to eliminate errors, this paper uses the SLERP algorithm [40]. The traditional quaternion interpolation is susceptible to issues such as Gimbal Lock, and SLERP, which involves quaternion calculations, can solve this problem and avoid the singularities, ensuring the smoothest path and the highest possible speed for the tool path.

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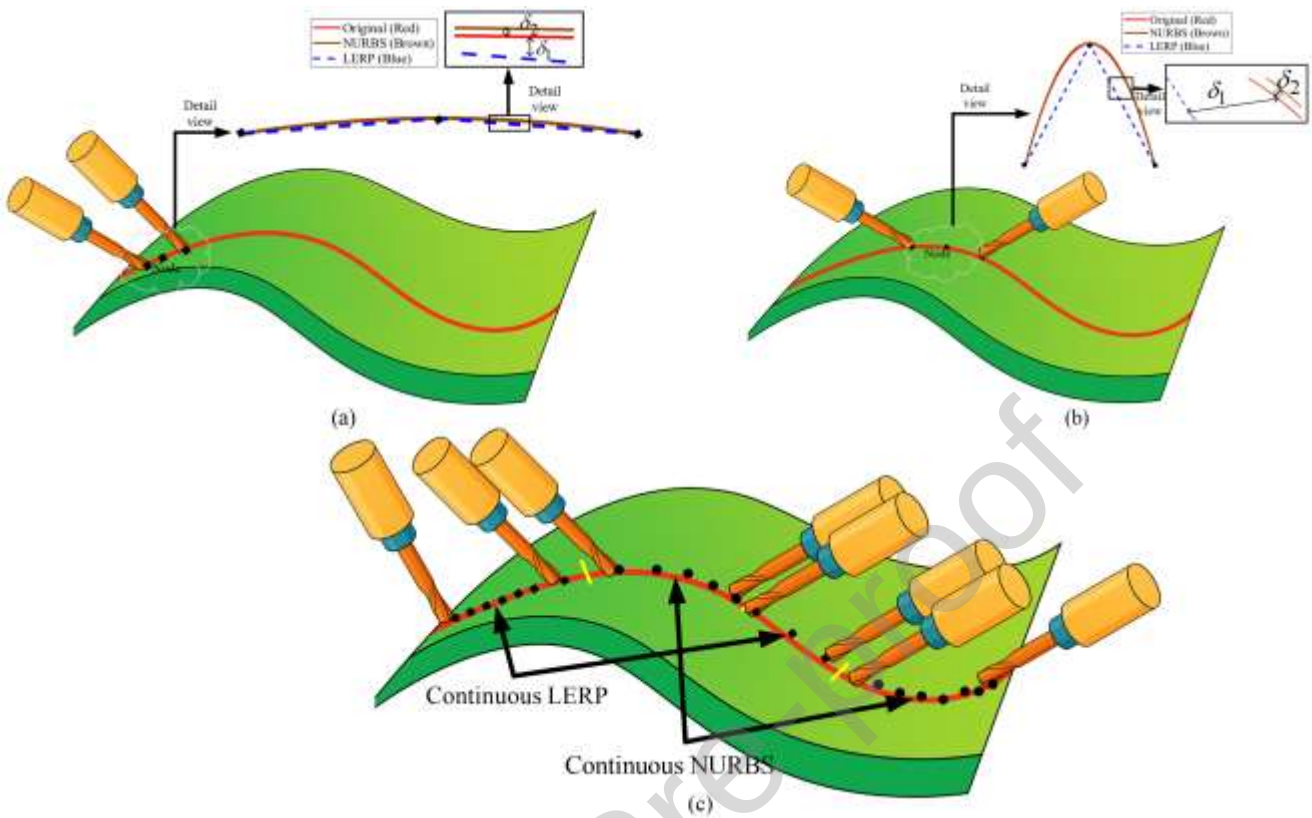

Figure 4 Schematic of TCP adaptive hybrid interpolation strategy: (a) LERP vs. NURBS TCP interpolation under low curvature, (b) LERP vs. NURBS TCP interpolation under high curvature, (c) Schematic of continuous LERP and NURBS hybrid interpolation for TCP trajectory

A quaternion is a four-dimensional number that can represent a rotation in three-dimensional space, and it is often used to describe the orientation of a vector in space. A unit quaternion can be expressed as follows:

$$\begin{aligned} q &= \left[\cos\left(\frac{\theta}{2}\right), \sin\left(\frac{\theta}{2}\right)\boldsymbol{B}\right] \\ &= \left[\cos\left(\frac{\theta}{2}\right), B_x \sin\left(\frac{\theta}{2}\right), B_y \sin\left(\frac{\theta}{2}\right), B_z \sin\left(\frac{\theta}{2}\right)\right], \end{aligned} \tag{25}$$

where θ is the angle of rotation, and $\boldsymbol{B}$= [B_x, B_y, B_z] is the axis of rotation.

Before interpolation, the starting tool axis vector v_s and ending vector $v_e$must be converted to their corresponding quaternions, $q_s$and q_e. This can be achieved by calculating the rotation from a standard reference vector (e.g., the positive Z-axis, $\boldsymbol{k}$= [0,0,1]) to the target tool axis vector. For any given vector $\boldsymbol{v}$:

Rotation axis:

$$\begin{aligned} \boldsymbol{B} &= \frac{\boldsymbol{k}\times\boldsymbol{v}}{\|\boldsymbol{k}\times\boldsymbol{v}\|} = \frac{\left[0\cdot v_z - 1\cdot v_y \quad 1\cdot v_x - 0\cdot v_z \quad 0\cdot v_y - 0\cdot v_x\right]}{\sqrt{(-v_y)^2 + v_x^2 + 0^2}} \\ &= \left[\frac{-v_y}{\sqrt{v_x^2+v_y^2}}, \frac{v_x}{\sqrt{v_x^2+v_y^2}}, 0\right]. \end{aligned} \tag{26}$$

Rotation angle:

$$\theta = \arccos(\boldsymbol{k}\cdot\boldsymbol{v}) = \arccos\left(0\cdot v_x + 0\cdot v_y + 1\cdot v_z\right) = \arccos(v_z). \tag{27}$$

By substituting $\boldsymbol{B}$ and θ into the quaternion formula above, the corresponding quaternion representation can be obtained.

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With the start and end quaternions q_s and q_e, the SLERP algorithm can smoothly interpolate between them to find the intermediate orientation quaternion $q(u)$ for a parameter $u \in [0,1]$:

$$q(u) = \frac{\sin\left((1-u)\frac{\Omega}{2}\right)}{\sin\left(\frac{\Omega}{2}\right)} q_s + \frac{\sin\left(u\frac{\Omega}{2}\right)}{\sin\left(\frac{\Omega}{2}\right)} q_e, \tag{28}$$

where Ω is the actual total angle of rotation of the tool axis in 3D space. It can be obtained using Eq. (12). Its relationship with the quaternion dot product is given by

$$\frac{\Omega}{2} = \arccos\left(\cos\left(\frac{\Omega}{2}\right)\right) = \arccos\left\{\max\left[-1, \min\left(1, q_s \bullet q_e\right)\right]\right\},$$

$$\begin{aligned} q_s \bullet q_e &= \cos\left(\frac{\theta_s}{2}\right)\cos\left(\frac{\theta_e}{2}\right) \\ &+ \frac{-J_s}{\sqrt{(-J_s)^2 + (I_s)^2}} \sin\left(\frac{\theta_s}{2}\right) \frac{-J_e}{\sqrt{(-J_e)^2 + (I_e)^2}} \sin\left(\frac{\theta_e}{2}\right) \\ &+ \frac{I_s}{\sqrt{(-J_s)^2 + (I_s)^2}} \sin\left(\frac{\theta_s}{2}\right) \frac{I_e}{\sqrt{(-J_e)^2 + (I_e)^2}} \sin\left(\frac{\theta_e}{2}\right). \end{aligned}$$

Eq. (28) guarantees that the interpolation path is the shortest spherical arc and that the angular velocity is constant.

This conversion is achieved through a standard quaternion rotation operation. An initial reference vector (typically the Z-axis, [0,0,1]), represented as a pure quaternion $q_0 = (0, 0, 0, 1)$, is rotated by the interpolated quaternion q_i by left-multiplying by q and right-multiplying by its inverse, q^{-1}. This yields the new tool axis vector, also represented as a pure quaternion, b_i:

$$b_i = q_i \otimes q_0 \otimes q^{-1}, \tag{29}$$

where, $\otimes$ represents quaternion multiplication. q_0 is the pure quaternion representing the initial reference direction. q^{-1} is the inverse of q_i. For a unit quaternion, its inverse is equal to its conjugate. If a quaternion is defined as $q_i = (g_i, u_i, v_i, w_i)$, its conjugate (and thus its inverse) is: $q^{-1} = (g_i, -u_i, -v_i, -w_i)$

The rotation operation described above requires performing quaternion multiplication twice. For any two quaternions, $q_a = (g_a, u_a, v_a, w_a)$ and $q_b = (g_b, u_b, v_b, w_b)$, the components of their product, $q_c = q_a \otimes q_b$, are calculated with Eq. (30):

$$\begin{cases} g_c = g_a g_b - u_a u_b - v_a v_b - w_a w_b, \\ u_c = g_a u_b + u_a g_b + v_a w_b - w_a v_b, \\ v_c = g_a v_b - u_a w_b + v_a g_b + w_a u_b, \\ w_c = g_a w_b + u_a v_b - v_a u_b + w_a g_b. \end{cases} \tag{30}$$

The final result of this calculation, b_i, is a pure quaternion with a scalar part equal to zero. Its vector (imaginary) part, $(I_i', J_i' K_i')$, corresponds to the new tool axis vector obtained after interpolation. By applying this process for a series of values for the parameter u from 0 to 1, the tool axis vectors for all intermediate points along the entire interpolation path can be obtained.

4.2.2 The Adaptive Hybrid Interpolation Strategy for TAV

A similar adaptive hybrid interpolation strategy for the TAV is proposed in this paper, which is also based on the trajectory curvature κ. The principle is illustrated in Figure 5. As shown in Figure 5(a), during small orientation changes caused by low-curvature paths, the deviation between the computationally efficient LERP and the ideal SLERP path is minimal.

However, as shown in Figure 5(b), when the trajectory enters a high-curvature region requiring a large rotation, LERP deviates significantly from the shortest spherical path, producing a notable pose error. Therefore, as shown in Figure 5(c), the final resulting TAV hybrid trajectory employs LERP in regions of gentle orientation change to enhance efficiency, and seamlessly switches to SLERP in regions of drastic orientation change to guarantee the smoothness and stability of the motion. Similar to the TCP strategy, the curvature-driven approach for TAV effectively mitigates swing error in high-risk regions. This completes the foundational framework of our adaptive method. The next section will now introduce the core innovation of this paper: a vector error coupling model that refines this foundational strategy by analyzing the geometric interplay between chord height and swing errors, enabling a more precise and efficient decision-making process.

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4.3 Refinement: the Vector Error Coupling Model and Final Decision Logic

The foundational adaptive strategies detailed in Sections 4.1 and 4.2 provide a robust framework for balancing machining efficiency and precision. By using local path curvature (κ) as the primary decision criterion, the system can effectively switch between high-efficiency LERP and high-precision NURBS/SLERP. However, a critical assumption in this foundational model is that high curvature universally translates to high error risk, thus always mandating a switch to the high-precision mode.

A deeper analysis is presented in this paper, revealing an opportunity for a more sophisticated and efficient approach. The final geometric accuracy at the tool tip is not merely a function of individual error magnitudes but is critically determined by the vectorial interaction between the TCP's chord height error and the TAV's swing error. This interaction, or coupling, can either amplify the total error or, under certain geometric conditions, significantly reduce it.

This section introduces the core innovation of this paper: a Vector Error Coupling Model that precisely describes this phenomenon. We then use this model to refine the foundational curvature-driven strategy into a more intelligent, dual-check decision logic that responds to the true, combined error risk.

4.3.1 The Vector Error Coupling Model

To understand the combined effect of interpolation errors, we must consider them as vectors. The total effective error vector at the tool tip, $\boldsymbol{E}_{\mathbf{total}}$, is the vectorial sum of the chord height error vector (δ) and the swing error displacement vector ($\boldsymbol{E}_{\mathbf{swing}}$):

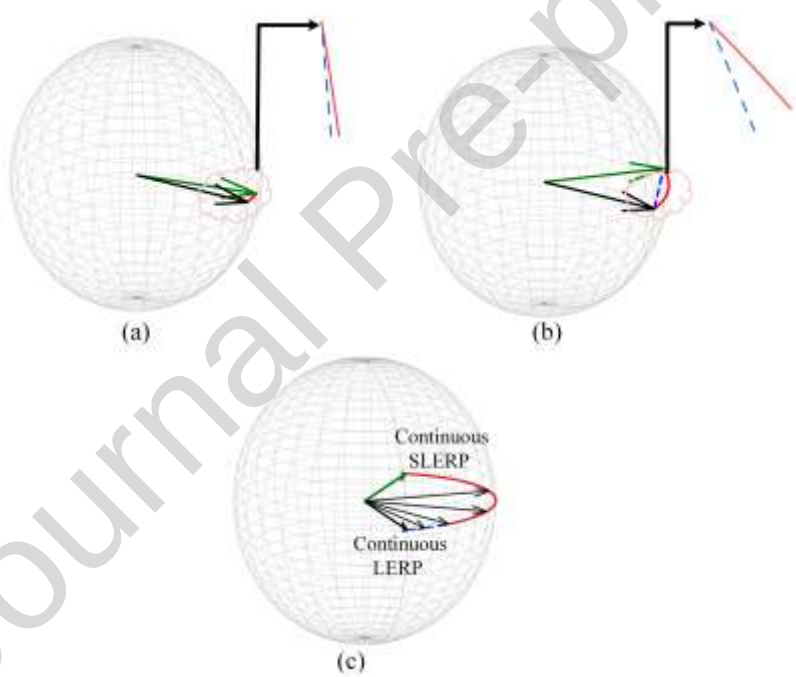

Figure 5 Schematic of the adaptive hybrid interpolation strategy for the TAV: (a) LERP vs. SLERP TAV interpolation under low curvature, (b) LERP vs. SLERP TAV interpolation under high curvature, (c) Continuous hybrid interpolation

$$\boldsymbol{E}_{\mathbf{total}} = \delta + \boldsymbol{E}_{\mathbf{swing}}. \tag{31}$$

The magnitude of this total error, which dictates the final machining accuracy, is given by:

$$\| \boldsymbol{E}_{\mathbf{total}} \| = \sqrt{\| \delta \|^2 + \| \boldsymbol{E}_{\mathbf{swing}} \|^2 + 2 \| \delta \| \cdot \| \boldsymbol{E}_{\mathbf{swing}} \| \cos(\theta)}, \tag{32}$$

where θ is the angle between the two error vectors. To implement this model, the magnitudes and directions of the individual error vectors for the current segment (assuming LERP+LERP interpolation) must be predicted:

Chord Height Error vector (δ):

The magnitude of the chord height error, $\| \delta \|$, for the local path segment, is calculated using Eq. (11), based on the segment's chord length and local curvature radius.

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The direction of δ is defined by the unit normal vector $\boldsymbol{n}_{\text{path}}$, which is perpendicular to the chord and points towards the ideal path. For three consecutive tool path nodes $\boldsymbol{A}_{i-1}$, $\boldsymbol{A}_i$, and $\boldsymbol{A}_{i+1}$(as depicted in Figure 2), the direction of $\boldsymbol{n}_{\text{path}}$ at node $\boldsymbol{A}_i$ can be reliably derived. First, define the chord vector connecting $\boldsymbol{A}_{i-1}$ to $\boldsymbol{A}_{i+1}$ as $\boldsymbol{C}_{i-1,\,i+1} = \boldsymbol{A}_{i+1} - \boldsymbol{A}_{i-1}$. The midpoint of this chord is $\boldsymbol{M}_i = (\boldsymbol{A}_{i-1} + \boldsymbol{A}_{i+1}) / 2$. The vector $\boldsymbol{D}_{Ai}$ from the chord midpoint M_i to the actual path node $\boldsymbol{A}_i$ inherently captures the deviation and points towards the arc, as shown by δ in Figure 2. Thus, the direction vector for δ is:

$$\boldsymbol{D}_{Ai} = \boldsymbol{A}_i - \boldsymbol{M}_i = \boldsymbol{A}_i - \frac{\boldsymbol{A}_{i-1} + \boldsymbol{A}_{i+1}}{2}. \tag{33}$$

The unit path normal vector $\boldsymbol{n}_{\text{path}}$ is then simply the normalized version of this direction vector:

$$\boldsymbol{n}_{\text{path}} = \frac{\boldsymbol{D}_{A_i}}{\| \boldsymbol{D}_{A_i} \|}. \tag{34}$$

The chord height error vector is then:

$$\delta = \| \delta \| \cdot \boldsymbol{n}_{\text{path}}. \tag{35}$$

Swing Error Displacement Vector ($\boldsymbol{E}_{\text{swing}}$):

The magnitude, $\| \boldsymbol{E}_{\text{swing}} \|$, representing the tool tip displacement, is estimated using Eq. (14), based on the tool length L and the dimensionless swing error ε.

The direction of $\boldsymbol{E}_{\text{swing}}$ (denoted by the unit vector $\boldsymbol{n}_{TAV}$ represents the maximum deviation of the LERP-interpolated TAV from the ideal SLERP-interpolated TAV at the tool tip. This direction represents the deviation trend of the tool axis. Specifically, it is determined by predicting the geometric deviation at the midpoint of the interpolation segment (u=0.5), where the error is maximum. $\boldsymbol{n}_{\text{TAV}}$ is calculated as the normalized vector difference between the ideal SLERP vector and the LERP vector at this midpoint, pointing towards the ideal trajectory.

$$\boldsymbol{E}_{\text{swing}} = \| \boldsymbol{E}_{\text{swing}} \| \cdot \boldsymbol{n}_{\text{TAV}}. \tag{36}$$

The angle θ between δ and $\boldsymbol{E}_{\text{swing}}$ is determined by their dot product:

$$\cos(\theta) = \frac{\delta \cdot \boldsymbol{E}_{\text{swing}}}{\| \delta \| \cdot \| \boldsymbol{E}_{\text{swing}} \|}. \tag{37}$$

The value of $\cos\theta$ directly reveals the error coupling mode, which is fundamentally correlated with the local surface geometry's concavity or convexity. Figure 6 visually illustrates this critical coupling effect.

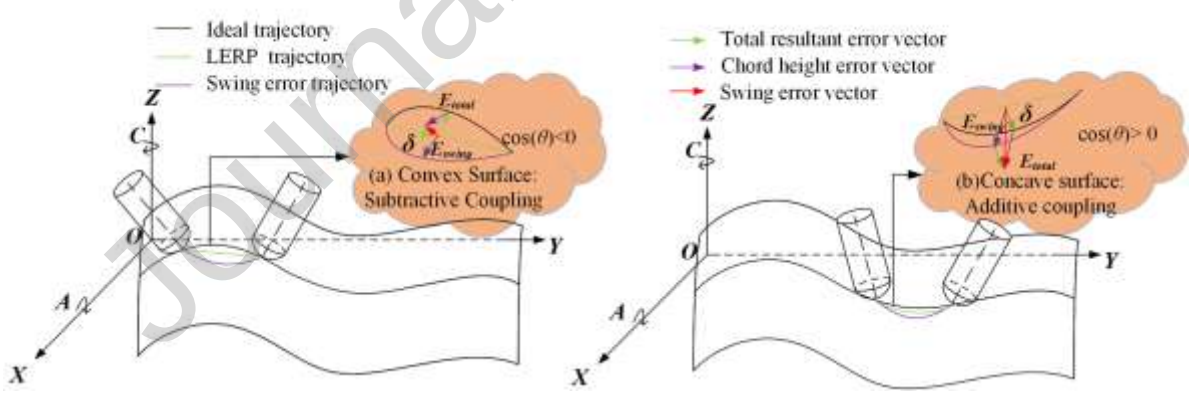

Figure 6 Effect of local surface concavity/convexity on error coupling

In five-axis machining, the bending direction of the tool path $\boldsymbol{n}_{\text{path}}$ and the direction of tool axis variation ($\boldsymbol{n}_{\text{TAV}}$) exhibit specific relative relationships due to the concavity or convexity of the workpiece surface. This geometric relationship directly influences the interaction of the two error vectors, δ and $\boldsymbol{E}_{\text{swing}}$:

Additive coupling: when $\cos\theta > 0$ (i.e., θ is an acute angle).

This typically occurs when machining a concave surface. In this scenario, the chord height error vector δ (pointing towards the path's center of curvature) and the swing error vector $\boldsymbol{E}_{\text{swing}}$ (tending to align with the tool's inclination towards the

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interior of the curve to avoid collision) have directions that are largely congruent. Their constructive superposition leads to an amplified total error $\| E_{\text{total}} \|$, as shown in Figure 6.

Subtractive coupling: when $\cos\theta < 0$ (i.e., θ is an obtuse angle).

This typically occurs when machining a convex surface. In this scenario, the directions of the chord height error vector δ and the swing error vector $\boldsymbol{E}_{\text{swing}}$ tend to oppose each other. Their mutual cancellation results in a reduced total error $\| E_{\text{total}} \|$, as depicted in Figure 6.

4.3.2 The refined dual-check adaptive strategy

By integrating the vector error coupling model with the foundational curvature-driven strategy, we formulate a refined Dual-Check Adaptive Strategy. This strategy does not discard the curvature criterion but uses it as a first-pass filter, followed by a more precise risk assessment based on error coupling for high-curvature segments. The refined decision logic is as follows. It is illustrated in the flowchart in Figure 7. The threshold is a theoretical value derived from the chord height error model (Eq. (11)). It corresponds to the critical curvature where the TCP chord height error (δ) alone reaches the preset tolerance ($\| E_{\text{total}} \|$). However, unlike traditional methods, exceeding this threshold does not force an immediate switch; instead, it serves as a trigger to activate the secondary error coupling check (via $\cos\theta$) to determine if the total error remains within limits due to the subtractive coupling effect.

Low Curvature Regions ($\kappa <$ threshold):

Decision: The foundational strategy is applied without modification. Both TCP and TAV use LERP interpolation.

Rationale: In these regions, individual error magnitudes are inherently low. The total error will remain within tolerance regardless of the coupling mode.

High Curvature Regions ($\kappa >$ threshold):

The system performs a secondary check at the current node by calculating $\cos\theta$ (Eq. (37)) to determine the error coupling mode.

- Case A: High curvature + additive coupling ($\cos\theta > 0$, corresponding to concave surface machining)

Analysis: In this scenario, δ and $\boldsymbol{E}_{\text{swing}}$ form an acute angle, leading to an amplified total error $\| E_{\text{total}} \|$. High curvature already magnifies individual error magnitudes, and additive coupling further exacerbates the total error. This constitutes a high total error risk, almost certainly exceeding the predefined tolerance.

Decision: To ensure machining precision, the system must switch to a high-precision hybrid interpolation strategy: NURBS interpolation for the TCP path and SLERP for the TAV orientation.

- Case B: High curvature + subtractive coupling ($\cos\theta \leq 0$, corresponding to convex surface machining)

Analysis: In this scenario, δ and $\boldsymbol{E}_{\text{swing}}$ form an obtuse angle, leading to a reduction in total error due to mutual cancellation. However, even with this beneficial cancellation, the resulting total error $\| E_{\text{total}} \|$ (calculated using Eq. (32)) must be explicitly checked against the predefined tolerance ($E_{\text{tolerance}}$) to ensure accuracy.

Sub-Decision 1: If $\| E_{\text{total}} \| \leq E_{\text{tolerance}}$:

Decision: Leveraging this advantageous error cancellation effect, the system can maintain LERP for TCP and LERP for TAV, even for high-curvature regions. This approach preserves computational efficiency without compromising the required machining accuracy, demonstrating a significant advantage over purely curvature-driven strategies.

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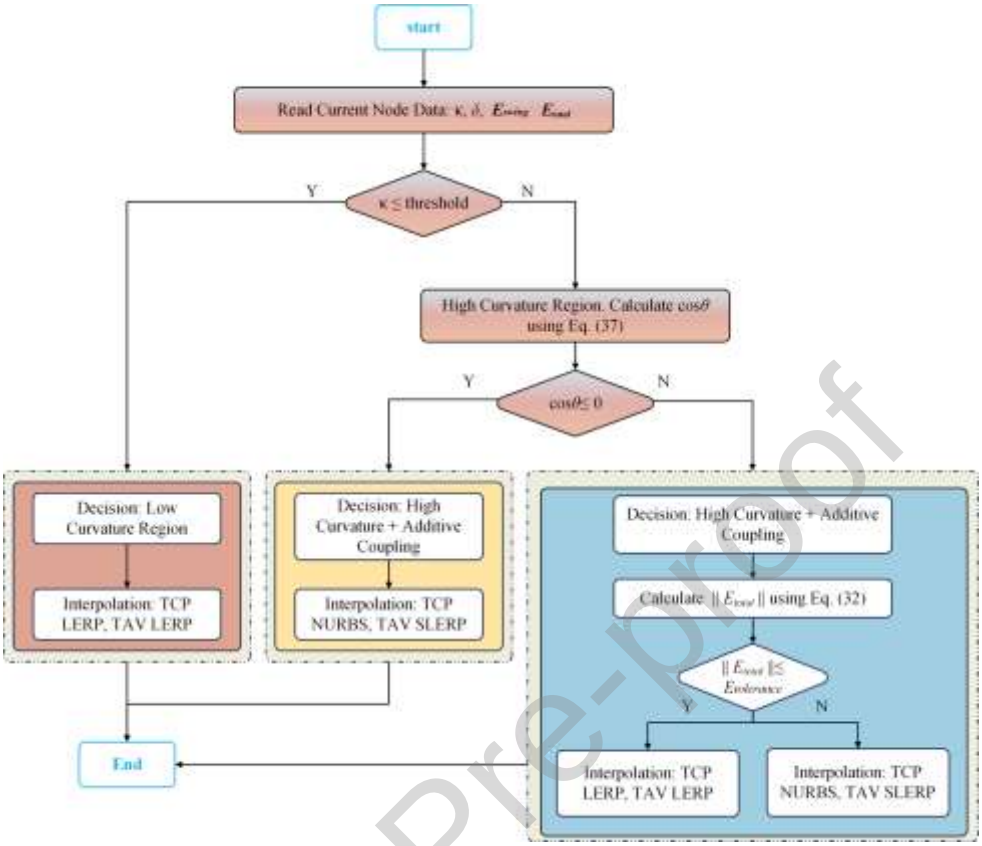

Figure 7 Flowchart of the refined dual-check adaptive interpolation strategy

Sub-Decision 2: If ‖ E_{total} ‖ > $E_{tolerance}$:

Decision: Despite the subtractive coupling, if the predicted total error still exceeds the tolerance, precision must be prioritized. The system must switch to a high-precision hybrid interpolation strategy: NURBS for TCP and SLERP for TAV.

This dual-check strategy, by integrating both curvature-based pre-screening and a sophisticated error-coupling-based risk assessment in the node space, facilitates a truly adaptive and intelligent control system. It dynamically balances machining efficiency and precision by directly evaluating the true error risk at the tool tip, leading to optimized performance across diverse surface geometries.

4.4 Overall algorithm framework

The overall framework of the adaptive hybrid interpolation algorithm proposed in this paper is illustrated in Figure 8. The algorithm's workflow is divided into three main stages.

The first stage, initialization and pre-processing, involves receiving the raw tool path data, consisting of a series of TCPs and their corresponding TAVs. The key task in this stage is coordinate system unification, where data that may originate from the Machine Coordinate System (MCS) is converted to the Workpiece Coordinate System (WCS) using the inverse kinematic model (Eqs. (3), (4)).

The second stage is the core adaptive interpolation loop, which iteratively processes the tool path. In this loop, the "three-point method" (Eqs. (5)-(8)) is used for geometric analysis to calculate the key geometric criterion in real-time: the local curvature κ. Based on a primary comparison of this curvature with a preset threshold, the system makes an intelligent initial decision:

For Low-Curvature Regions (κ<threshold): The system enters a high-efficiency mode, employing simple LERP for both the TCP and TAV (Eq. (17)) to maximize machining efficiency. In these regions, individual interpolation errors are inherently low and remain within tolerance.

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For High-Curvature Regions (κ>threshold): A secondary, more precise risk assessment is performed based on the error coupling mode. The system calculates the cosine of the angle θ between the chord height error vector (δ) and the swing error displacement vector ($\boldsymbol{E}_{\mathbf{swing}}$) using the vector error coupling model (Eq. (37)).

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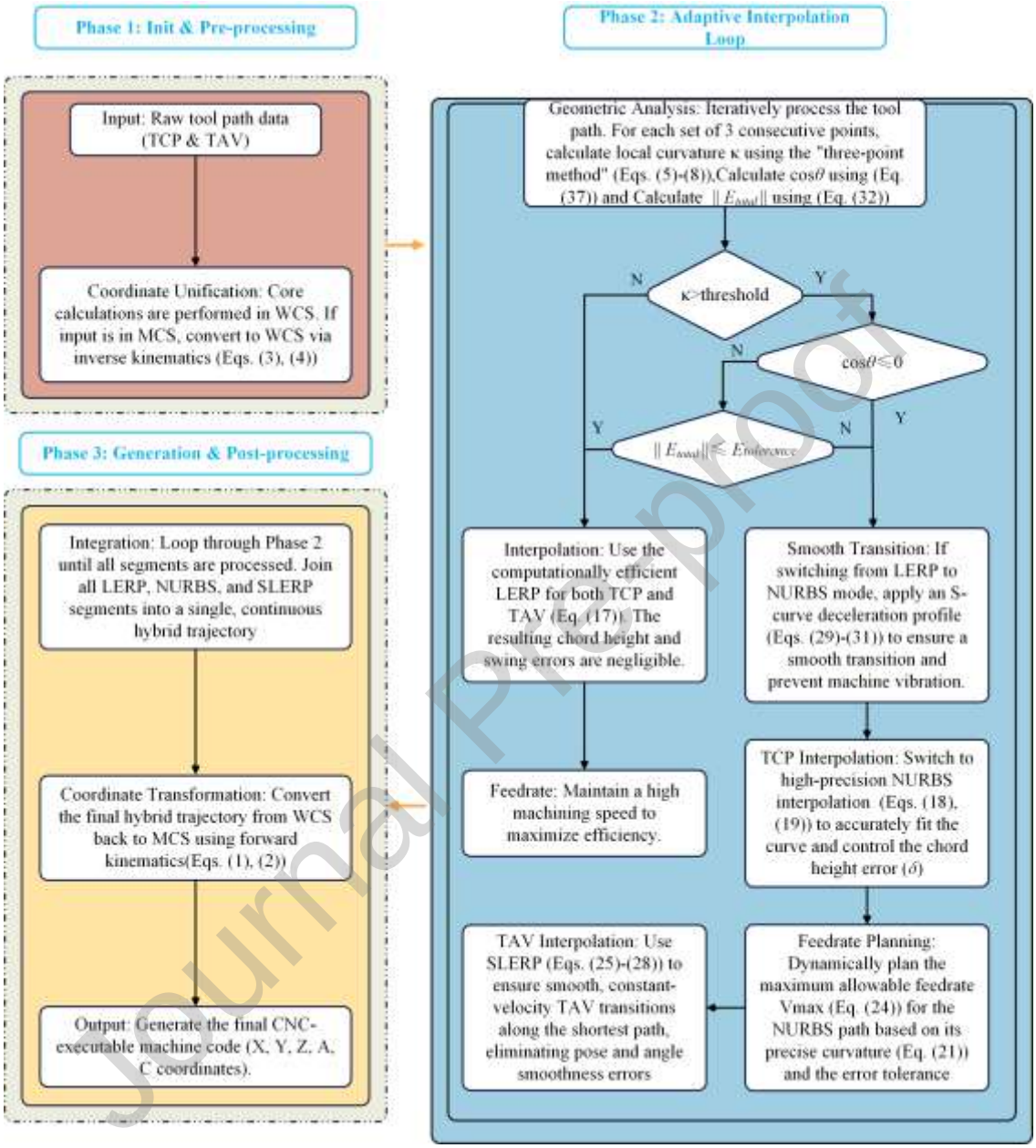

Figure 8 Overall framework of the adaptive hybrid interpolation algorithm

Case A: Additive Coupling ($\cos\theta > 0$, typically corresponding to concave surface machining): If an additive coupling mode is detected, indicating that the two error vectors tend to amplify each other, the total error risk is high and likely to exceed the predefined tolerance. To ensure machining precision, the system switches to a high-precision mode where the path and orientation are decoupled: the TCP is precisely interpolated using NURBS (Eqs. (18), (19)) to strictly control contour error (δ), while the TAV is smoothly interpolated using SLERP (Eqs. (25)-(28)) to eliminate swing error ($\boldsymbol{E}_{\mathbf{swing}}$).

Case B: High Curvature + Subtractive Coupling ($\cos\theta \leq 0$, typically corresponding to convex surface machining): If a subtractive coupling mode is detected, indicating that the two error vectors tend to cancel each other out, the system then performs a final check: it calculates the magnitude of the total error $\| E_{\text{total}} \|$ (Eq. (32)) and compares it against the predefined tolerance ($E_{\text{tolerance}}$).

If $\| E_{\text{total}} \| \leq E_{\text{tolerance}}$: The accuracy requirement is met. The system maintains LERP for both the TCP and TAV to preserve

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computational efficiency, as the accuracy requirement is met even with high curvature.

If $\| E_{\text{total}} \| > E_{\text{tolerance}}$: Despite the subtractive coupling effect, if the predicted total error still exceeds the tolerance, precision must be prioritized. The system switches to the high-precision mode, employing NURBS for TCP and SLERP for TAV.

The third stage involves the execution of the selected interpolation methods and generation of control commands for the CNC machine.

This refined dual-check adaptive strategy, by integrating both curvature-based pre-screening and a sophisticated error-coupling-based risk assessment in the node space, facilitates a truly adaptive and intelligent control system. It dynamically balances machining efficiency and precision by directly evaluating the true error risk at the tool tip, leading to optimized performance across diverse surface geometries.

5 Simulation and experiment

To validate the effectiveness and performance of the proposed hybrid interpolation algorithm, a series of simulations was conducted. Figure 9 illustrates the 3D model of the complex freeform surface used as the test workpiece in this study. The surface is characterized by continuously varying curvature, providing suitable conditions to evaluate the interpolation algorithm's performance. For the machining process, a ball-end mill with a radius of 2 mm and a tool length (L) of 20 mm was selected. To meet the stringent surface integrity requirements typical of high-precision finishing operations, such as aero-engine blades or precision molds, a maximum allowable chord height error of 3 μm was specified as the user-defined tolerance. For a detailed comparative analysis, two sets of three consecutive points were sampled from the toolpath on this surface: one set from a region of high curvature and another from a region of low curvature, as detailed in Table 1.

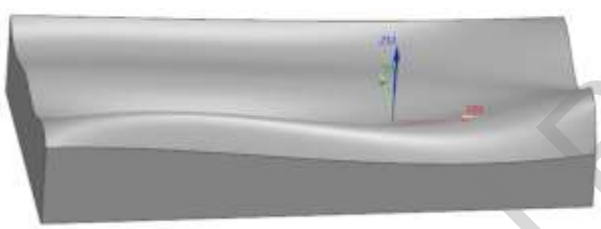

Figure 9 Test workpiece model

To evaluate the performance of the proposed hybrid interpolation method under varying geometric conditions, Figure 10 illustrates the interpolation results obtained using LERP and NURBS in two distinct regions of the toolpath: one with high curvature and one with low curvature.

Table 1 Coordinates of sampled points in high and low curvature regions

No.	x(mm)	y(mm)	z(mm)	I	J	K
1	-32.423	-10.348	8.101	0.353	-0.833	0.425
2	-31.866	-10.735	8.066	0.355	-0.879	0.318
3	-30.316	-10.974	8.000	0.356	-0.901	0.246
4	-10.692	-11.175	6.188	0.361	-0.917	0.171
5	-9.217	-11.117	6.028	0.362	-0.913	0.190
6	-7.747	-11.055	5.872	0.361	-0.908	0.211

In the high-curvature region (Figure 10(a)), the LERP method exhibits a clear deviation from the ideal trajectory, particularly near areas with rapid changes in curvature. This results in a maximum interpolation error of approximately 0.0189 mm. In contrast, the NURBS interpolation demonstrates significantly improved fidelity to the reference path, reducing the maximum error to 0.0017 mm. The smoothness and continuity provided by NURBS are evident in its ability to closely track the ideal trajectory throughout the segment.

In the low-curvature region (Figure 10(b)), both interpolation schemes perform better due to the relatively mild geometric variation. However, NURBS still achieves a notably higher precision, with a maximum error of only 0.000041 mm, compared to 0.000744 mm observed in the LERP result.

These results confirm that interpolation errors become more pronounced with increasing curvature when using linear schemes, while NURBS-based methods maintain a high level of accuracy and continuity. This reinforces the suitability of

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incorporating NURBS into the hybrid interpolation strategy, particularly in precision machining of freeform surfaces where trajectory smoothness is critical.

To evaluate the orientation interpolation performance within the proposed trajectory planning framework, Figure 11 compares LERP and SLERP for TAV transitions under both high and low curvature conditions on the unit quaternion sphere.

In the high-curvature scenario illustrated in Figure 11(a), three consecutive key orientation points are sampled and visualized as green markers on the unit sphere. The LERP interpolation (black dashed line), which linearly interpolates quaternion components in Euclidean space, leads to a path that deviates from the spherical surface, thereby violating the unit norm constraint of quaternions. This deviation becomes more evident in the magnified view, where the actual interpolation path (magenta line) diverges from the ideal spherical trajectory (red line), indicating geometric inaccuracy and potential issues in tool axis continuity. Such distortion can lead to abrupt changes in orientation during machining, affecting surface quality and tool wear.

In contrast, SLERP produces a smooth and geometrically correct interpolation (red solid line) that remains strictly on the unit sphere, preserving the quaternion norm and ensuring consistent tool orientation. The actual implementation (magenta line) closely follows the ideal SLERP path, confirming both numerical stability and high accuracy of the method.

As shown in Figure 11(b), the interpolation performance gap between LERP and SLERP narrows in the low-curvature case due to the minimal angular displacement between orientations. However, even in this case, the SLERP method continues to maintain better geometric consistency, while LERP exhibits a slight deviation from the ideal manifold.

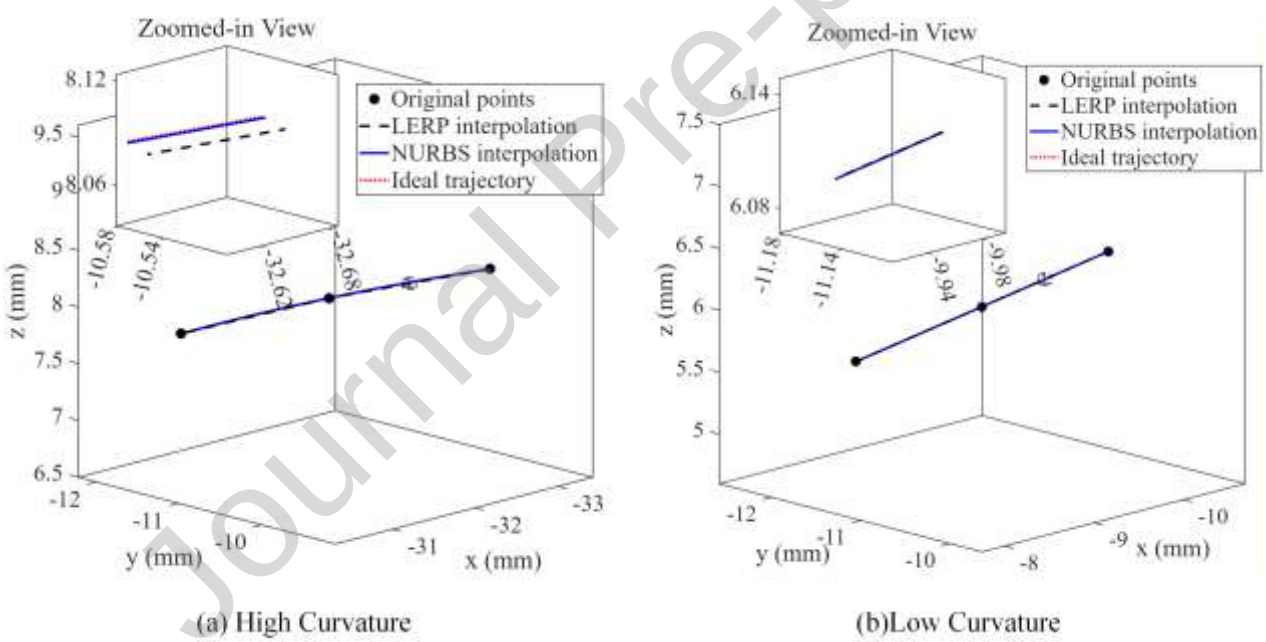

Figure 10 Comparison of LERP and NURBS for TAP interpolation under different curvature conditions

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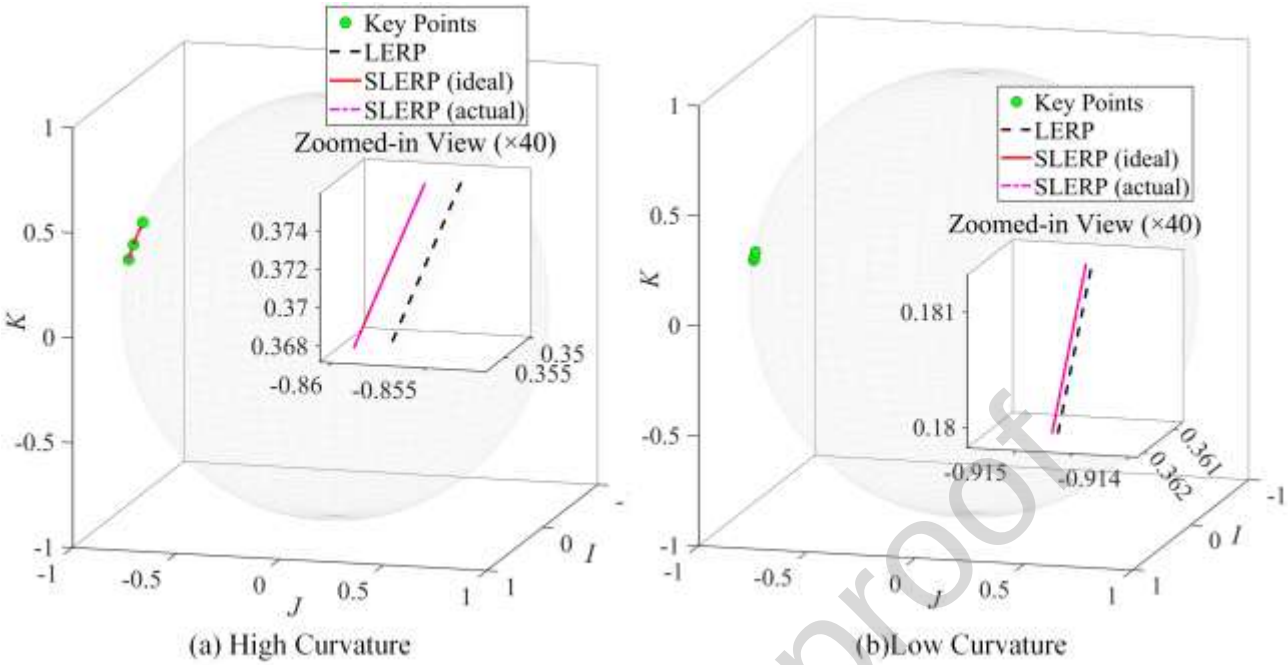

Figure 11 Comparison of LERP and SLERP for TAV interpolation under different curvature conditions

These observations affirm that SLERP offers significant advantages in high-curvature orientation transitions by ensuring rotation fidelity and smoothness. Incorporating SLERP into the hybrid interpolation strategy contributes to improved tool axis stability, particularly in the machining of geometrically complex freeform surfaces.

To further investigate the trade-off between interpolation accuracy and computational efficiency, Figure 12 presents a comparison of angular change and computation time for SLERP and LERP methods across high- and low-curvature regions.

In the high-curvature scenario (Figure 12(a)), SLERP maintains a consistent angular change between successive interpolation points, ensuring smooth and geometrically correct orientation transitions. LERP, while faster in terms of computation time by approximately one order of magnitude, exhibits slight fluctuations in angular change due to its deviation from the unit quaternion manifold, consistent with the pose error trends shown in Figure 12.

In the low-curvature region (Figure 12(b)), the angular changes produced by both methods become nearly identical, and the pose error introduced by LERP, as previously shown, becomes negligible. Importantly, LERP maintains a significant advantage in computation time under both curvature conditions.

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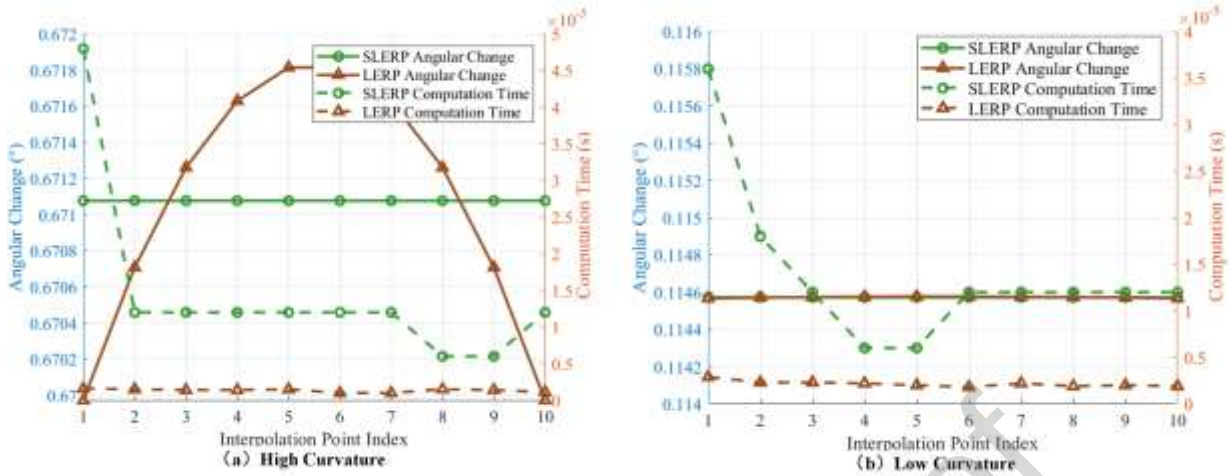

Figure 12 Comparison of angular change and computation time between SLERP and LERP under different curvatures

These results suggest that while SLERP is indispensable in high-curvature segments to preserve orientation accuracy, LERP may serve as a computationally efficient alternative in low-curvature regions, where its accuracy degradation is minimal. This observation reinforces the effectiveness of a curvature-aware hybrid interpolation strategy, where SLERP and LERP are selectively applied based on local geometric complexity.

To evaluate the overall effectiveness of the proposed hybrid interpolation strategy along the full machining trajectory, Figure 13 provides a point-wise comparison of local curvature, chord height error, and computation time along a linear toolpath consisting of 30 interpolation points.

As shown in Figure 13(a), the curvature profile of the tool tip path varies significantly along the trajectory, with high curvature concentrated at the beginning and end segments, and a relatively flat midsection. This non-uniform curvature distribution serves as a representative test case for evaluating curvature-adaptive interpolation schemes. Figure 13(b) compares the chord height errors generated by LERP, NURBS, and the proposed hybrid method. It is observed that LERP produces excessive errors up to 18 μm in regions with higher curvature, exceeding the allowable tolerance (3 μm, indicated by the magenta dashed line). NURBS interpolation consistently satisfies the error threshold but at the cost of increased computational load. The hybrid method successfully maintains the interpolation error within the tolerance across all segments by switching between LERP and NURBS depending on local curvature, thus achieving a good balance between precision and efficiency.

Figure 13(c) further confirms this balance by illustrating the corresponding computation times. While LERP remains the fastest method and NURBS the most computationally expensive, the hybrid method significantly reduces the average computation time compared to pure NURBS, particularly in low-curvature segments where LERP is employed without violating the error constraint.

These results collectively demonstrate the baseline performance of the curvature-aware hybrid interpolation strategy, which confirms that dynamically adjusting the interpolation scheme based on curvature is a viable approach.

To comprehensively validate the performance of the proposed hybrid interpolation strategy, Figure 14 compares the chord height error, pose error, and computation time across the full interpolation sequence, using LERP-only, SLERP-only, and hybrid methods.

As shown in Figure 14(a), the calculated chord height error of the tool tip path is used as the decision driver for selecting between LERP and NURBS in position interpolation. Regions where the chord error exceeds the threshold of 3 μm (indicated by the magenta dashed line) are identified as high-curvature segments requiring NURBS interpolation, while LERP is employed in low-curvature regions. This curvature-aware switching ensures that the hybrid method stays entirely within the geometric tolerance.

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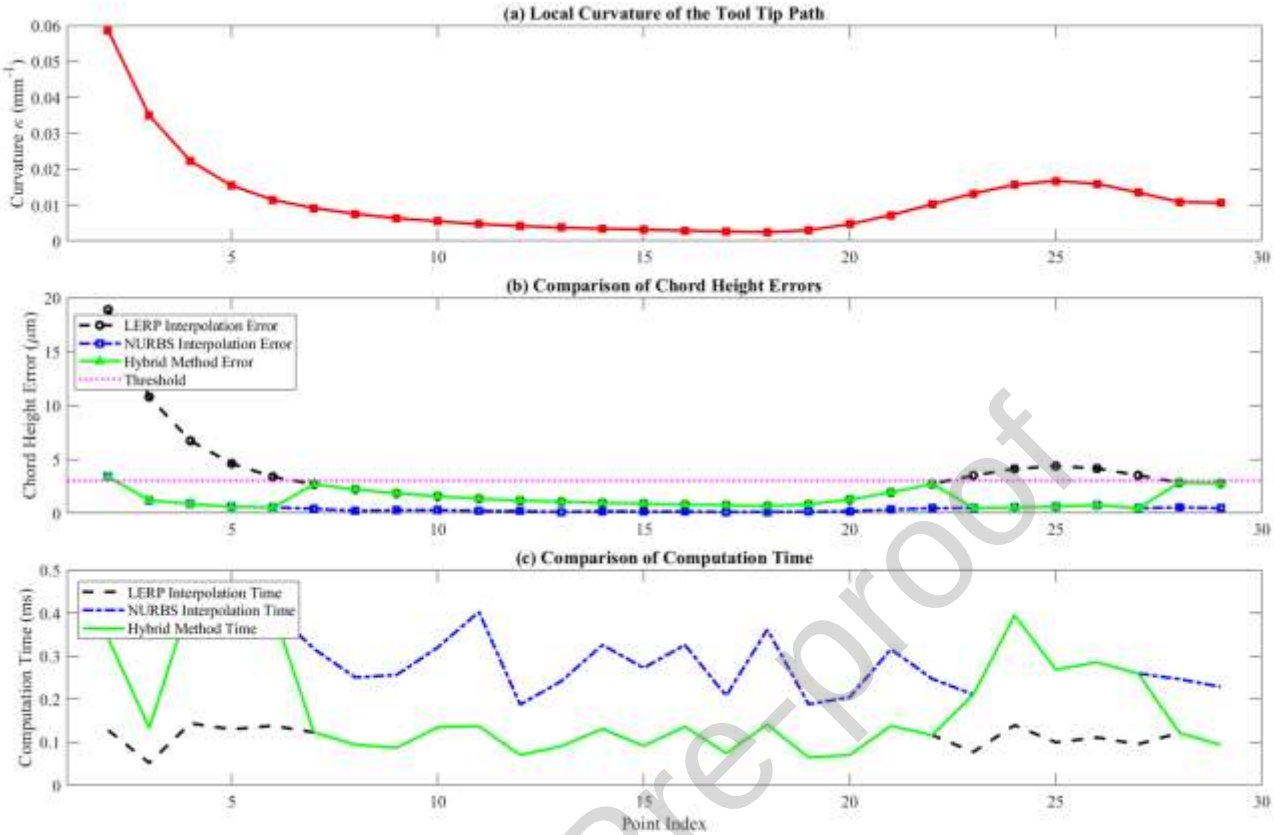

Figure 13 Comparison of interpolation accuracy and efficiency for LERP, NURBS, and hybrid methods

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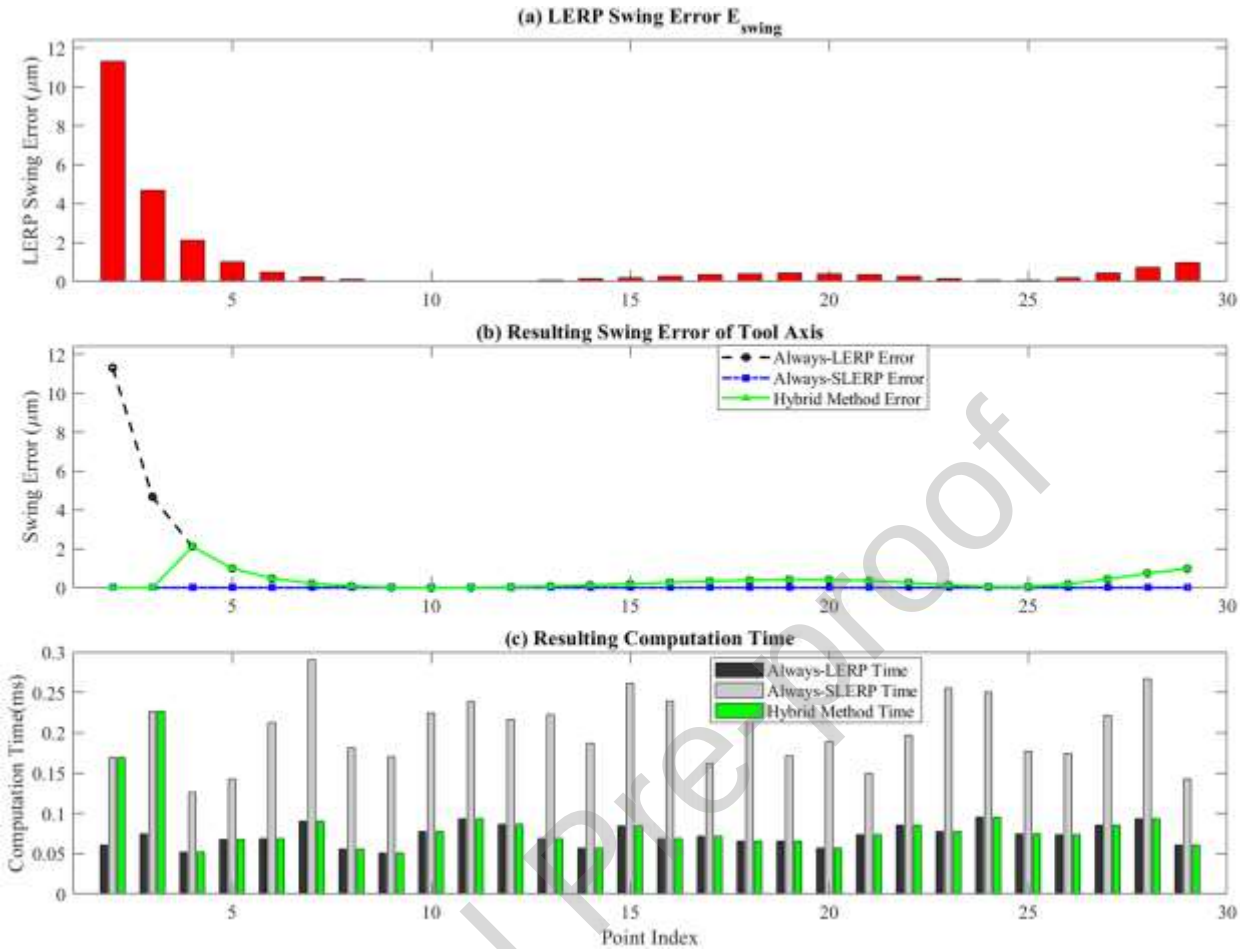

Figure 14 Error and efficiency comparison of LERP, SLERP, and hybrid interpolation

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The resulting pose errors of the tool axis for the three interpolation methods are presented in Figure 14(b). As expected, the LERP-only approach exhibits significantly higher pose error up to the order of 10^{-3} due to deviation from the unit quaternion constraint, particularly in regions where chord error is also high. SLERP-only interpolation effectively suppresses this error but at the cost of increased computation. The hybrid method, by selectively applying SLERP only where needed, maintains low pose error comparable to SLERP-only while avoiding unnecessary overuse.

The corresponding computation times are compared in Figure 14(c). While SLERP-only interpolation incurs the highest computational load, and LERP-only the lowest, the hybrid method achieves a substantial reduction in average computation time compared to SLERP, without compromising geometric or pose accuracy.

While Figures 13 and 14 demonstrate the effectiveness of the hybrid approach in managing individual error sources based on curvature-related thresholds, Figure 15 delves deeper by comparing the total geometric error ($\| E_{\text{total}} \|$) resulting from four distinct strategies, explicitly showcasing the superiority of incorporating the error coupling analysis.

The point-wise total geometric error for each strategy along the toolpath is presented in Figure 15(a). As expected, the Always LERP+LERP strategy (grey bars) exhibits significant errors, particularly in the initial and final segments (simulated high-risk regions), frequently exceeding the 3 μm tolerance line (magenta dashed line). Conversely, the Always NURBS+SLERP strategy (blue bars) maintains negligible error throughout, confirming its high-precision nature but implying higher computational cost.

The key comparison is between the Single Curvature Method (represented by the orange bars) and the proposed method. The 'Single Curvature Method' corresponds to the foundational strategy described in Sections 4.1 and 4.2. It relies solely on a rigid curvature threshold (κ> threshold) to trigger the high-precision NURBS+SLERP mode, without considering the

vector error coupling effect. The Single Curvature Method manages to control the error in many segments. However, critically, in the final segments of the path (approximately points 21 onwards), the total error produced by the Single Curvature Method exceeds the 3 μm tolerance. This failure occurs because, while curvature might be high in this region (triggering potentially a switch for TAV like in Fig 14), this simpler strategy fails to account for the additive coupling effect between the TCP chord height error (δ) and the TAV swing error ($\boldsymbol{E}_{\mathbf{swing}}$). As indicated by the simulated coupling factor in Figure 15(b), this end region corresponds to an additive coupling scenario ($\cos\theta \geq 0$, red shaded area), where the individual errors compound significantly.

In stark contrast, the proposed Hybrid (Dual-Check) method (green bars) successfully maintains the total geometric error strictly below the 3 μm tolerance across the entire toolpath, including the challenging end segments. This is achieved because the dual-check logic explicitly incorporates the error coupling analysis. Upon identifying a high-risk region (first check) and detecting an additive coupling mode ($\cos\theta \geq 0$, second check, as shown in Figure 15(b)), the strategy correctly mandates a switch to the high-precision NURBS+SLERP mode. By considering the vectorial interaction of errors, the dual-check method effectively prevents the error amplification that plagues the simpler strategies in additive coupling regions.

Therefore, Figure 15 compellingly validates the advantage of introducing the error coupling model (analyzing the $\cos\theta$ or concavity/convexity) into the adaptive strategy. The proposed Hybrid (Dual-Check) method demonstrates superior robustness and reliability by ensuring the final geometric accuracy remains within tolerance under all coupling conditions, unlike the Single Curvature Method which falters when error amplification due to additive coupling is ignored.

Taken together, these results demonstrate that the proposed hybrid strategy provides a balanced solution—preserving high interpolation fidelity in complex regions while minimizing computational overhead in flatter segments—making it highly suitable for real-time and precision-oriented CNC machining applications.

To visualize the spatial distribution of interpolation errors over a complex freeform surface, Figure 16 presents 3D error maps corresponding to four different interpolation strategies: (a) LERP, (b) the proposed hybrid method, (c) SLERP+NURBS, and (d) the reference method described in [3].

As shown in Figure 16(a), the LERP method yields significant surface deviation, especially in high-curvature boundary regions, where the local approximation error exceeds 0.025 mm. The error pattern reveals noticeable discontinuities and uneven distribution, indicating poor geometric fidelity for complex surfaces.

In contrast, the proposed method (Figure 16(b)) demonstrates a substantial improvement, with the error uniformly controlled across the surface and maximum deviations reduced to below 0.003 mm. This confirms the effectiveness of the proposed dual-check adaptive strategy in maintaining both positional and orientational continuity throughout the toolpath.

The SLERP+NURBS method (Figure 16(c)), which applies advanced interpolation in both position and orientation, also exhibits low error levels. However, it lacks the decision logic employed in the proposed method, leading to computational overhead in regions where simpler interpolation would suffice.

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Figure 15 Comparison of total geometric error for different strategies and simulated coupling mode

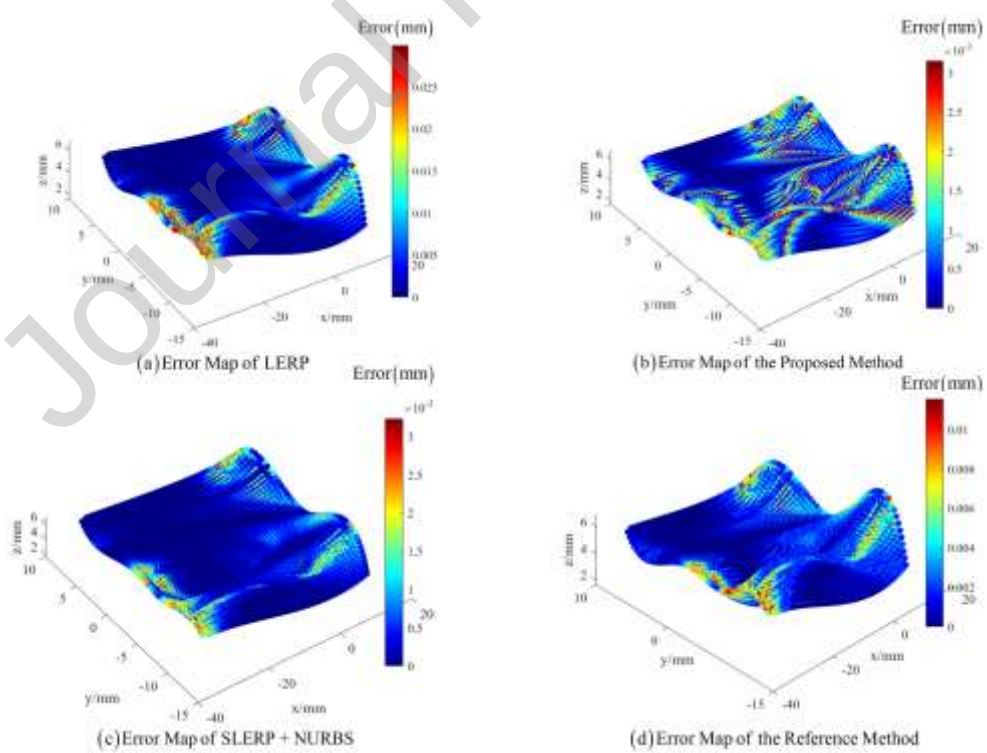

Figure 16 Comparison of error maps for LERP, proposed method, SLERP+NURBS, and reference method

The reference method from Ref. [3] (Figure 16(d)), although capable of suppressing peak deviations at specific points, focuses solely on mitigating oscillation-induced outliers. As a result, it neglects the majority of internal interpolation points,

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causing uneven error distribution across the surface. This limitation highlights the advantage of the present work, which provides a globally optimized trajectory by comprehensively addressing both end-point accuracy and continuity across the entire toolpath.

These findings validate the robustness and superiority of the proposed method for complex freeform surface machining, achieving both high accuracy and efficiency through adaptive interpolation control.

The simulated machining error surfaces generated using LERP and the proposed hybrid method are illustrated in Figure 17. As shown, LERP results in widespread surface deviations, with maximum errors exceeding 0.025 mm. These errors are particularly pronounced along sharp curvature transitions, reflecting the limitations of linear interpolation in maintaining smooth tool motion.

In contrast, the proposed method significantly reduces surface error, with deviations confined within 0.0036 mm across the entire region. The improved surface uniformity and reduced distortion confirm the effectiveness of the proposed dual-check adaptive strategy in practical machining scenarios.

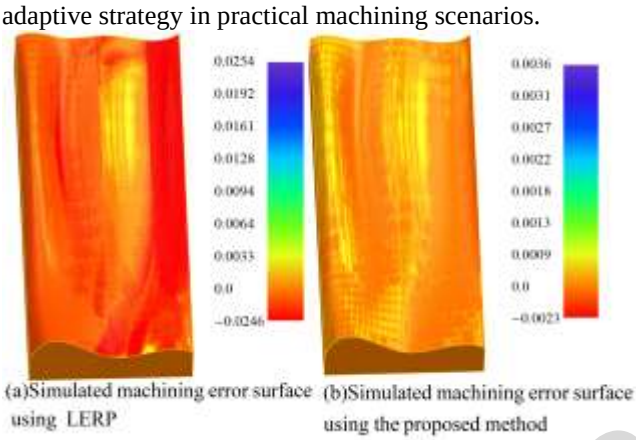

Figure 17 Comparison of simulated machining error surfaces

The five-axis CNC machining process simulated in VERICUT, using an A–C dual-turntable machine tool, is illustrated in Figure 18. The workflow includes rough and finish machining of a complex freeform surface.

As shown in Fig. 8(e) and (f), the proposed interpolation method produces a noticeably smoother surface compared to conventional linear interpolation. This confirms the practical effectiveness of the method in improving machining accuracy and surface quality.

To assess the real-time computational performance of different interpolation algorithms, Figure 19(a) shows a self-built CNC experimental platform designed specifically for interpolation efficiency testing. The platform integrates a GNS motion controller, servo drive system, and a five-axis machine tool structure, enabling flexible verification of algorithm execution time in an embedded environment. It serves as a general-purpose testbed rather than a replication of any specific industrial configuration.

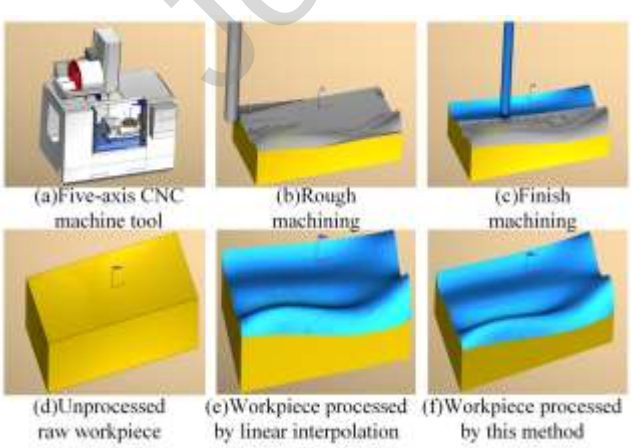

Figure 18 VERICUT Simulation of five-axis CNC machining process and workpiece comparison

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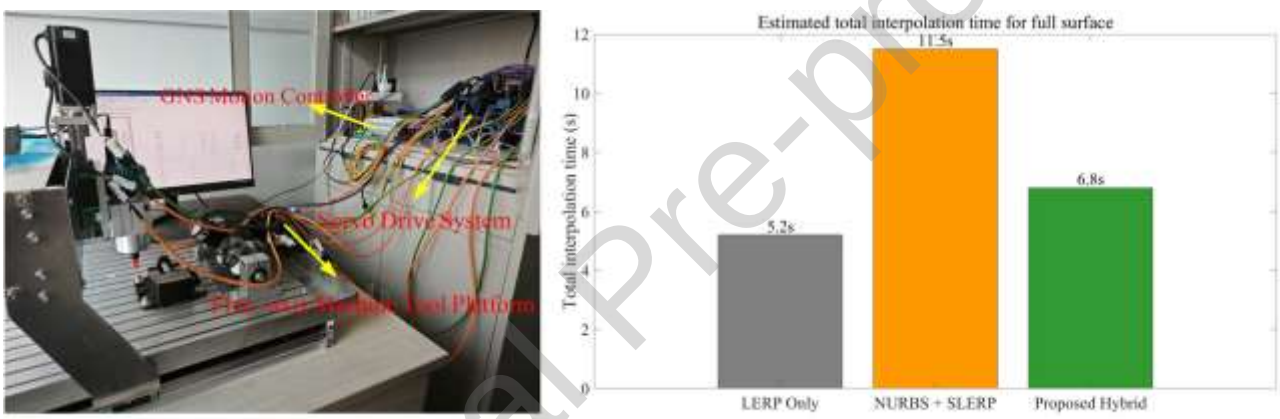
The estimated total interpolation time required to process the complete freeform surface is presented in Figure 19(b). To validate industrial feasibility, these benchmarks were conducted on a Googol GNS series embedded motion controller, equipped with a Quad-core 2.0 GHz processor, 2 GB RAM, and a gLink-II gigabit bus for high-speed synchronization.

As shown in Figure 19(b), the proposed hybrid method (6.8 s) significantly reduces the computational burden compared to the full NURBS+SLERP scheme (11.5 s). To address real-time feasibility, we further analyzed the absolute interpolation time per cycle. The efficient LERP mode averages approximately 0.062 ms (62 μs) per point, while the high-precision NURBS+SLERP mode averages 0.225 ms (225 μs). Crucially, the system's worst-case latency—occurring during complex mode switching—was recorded at 0.310 ms (310 μs).

Considering that the typical interpolation cycle of the GNS controller is set to 1 ms (1000 μs), the proposed algorithm's worst-case latency consumes only a fraction (approx.31%) of the time slot. This leaves ample computational margin for other tasks such as PLC logic and I/O processing. These results confirm that the method strikes a favorable balance between efficiency and precision, making it fully feasible for real-time implementation in embedded industrial CNC systems.

The Honyxi U200 five-axis CNC machine tool, shown in
Figure 20, features a dual-turntable design for enhanced flexibility and was used for the complete machining process of a workpiece. This system is equipped with high-precision spindles and advanced motion control, ideal for tasks such as freeform surface processing.



(a)Self - built CNC platform for interpolation efficiency testing (b)Estimated total interpolation time for full surface using different methods

Figure 19 Interpolation efficiency testing using self-built CNC platform and comparative timing results

The machining was performed using a ball-end mill with a radius of 2 mm and a tool length (L) of 20 mm, similar to the setup used in the simulation. The workpiece material is aluminum 6061, and the machining was conducted at a spindle speed of 8000 r/min. The images show the following steps: (a) the five-axis CNC machine tool setup, (b) the unprocessed raw workpiece, (c) rough machining, (d) workpiece after rough machining, (e) finish machining, and (f) the workpiece after finish machining.

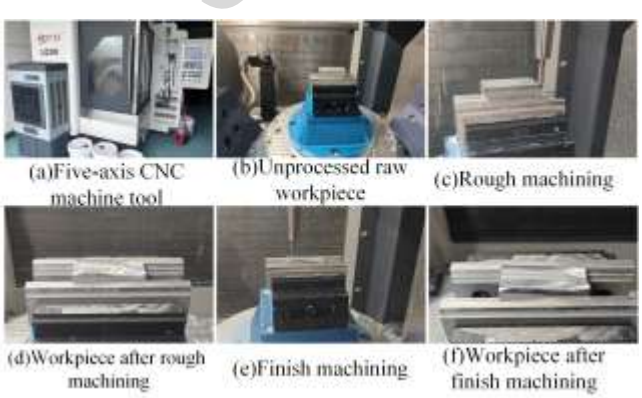


Figure 20 Machining process on the Honyxi U200 five-axis CNC machine tool

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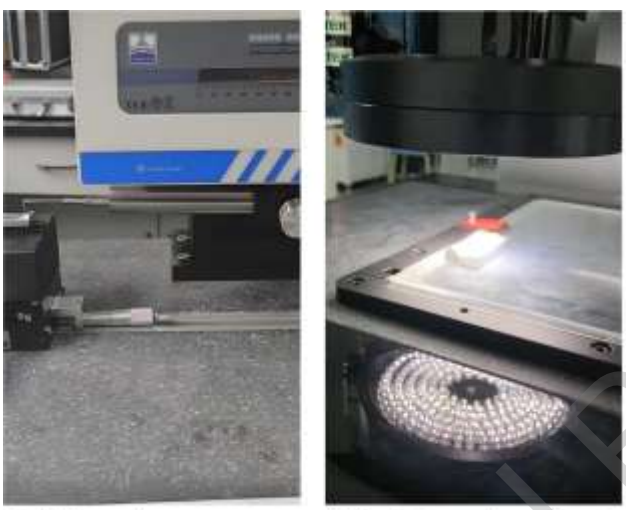
In this paper, the Tianzhun VME222 optical measurement system and the SRA-1 roughness gauge were used to evaluate machining accuracy and surface quality.

Figure 21(b) shows how the VME222 system captures surface deviations using non-contact 3D scanning. This method offers high precision and is used to compare machining errors from different interpolation techniques, validating the optimized interpolation method.

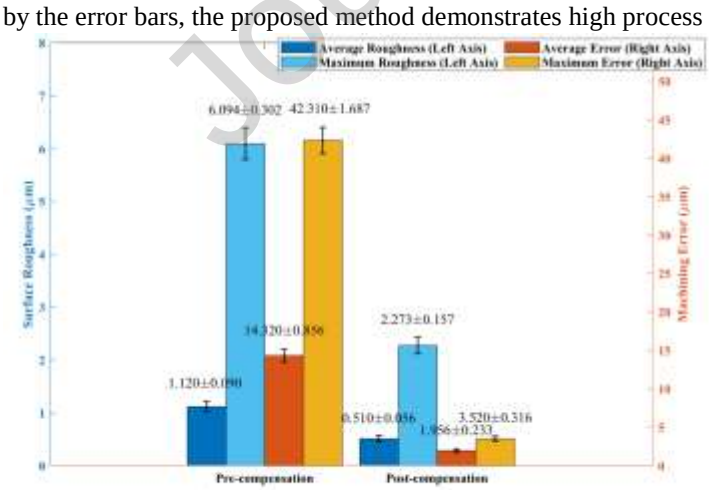
The measurement of surface roughness with the SRA-1, which uses a stylus to gather data and calculate roughness values, is demonstrated in Figure 21(a). The instrument was calibrated for aluminum 6061 to accurately measure fine surface details. Surface roughness (Ra) and maximum roughness (Rmax) were used to compare traditional linear interpolation and the optimized method.



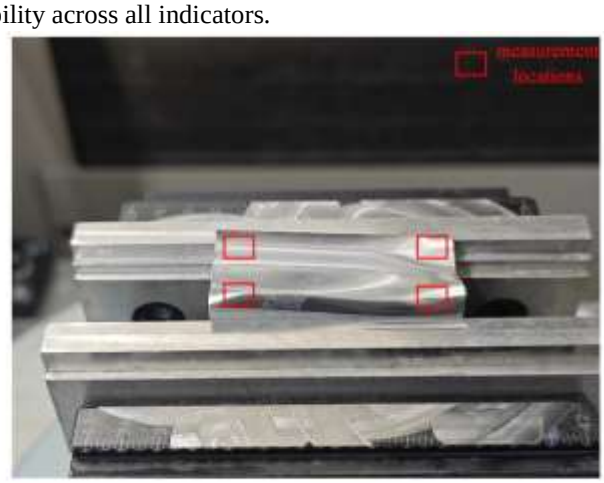
(a)Surface roughness measurement using a contact profilometer (b)Machining error inspection using an optical measurement system

Figure 21 Surface roughness and machining error measurement using different systems

A comprehensive statistical analysis of the machining experiments is presented in Figure 22, summarizing the results from three independent trials (*n*=3). Specifically, a detailed quantitative comparison of four key performance metrics between the traditional pre-compensation method and the proposed post-compensation strategy is provided in Figure 22(a). As indicated by the error bars, the proposed method demonstrates high process stability across all indicators.



(a)Comparison of roughness and error: Pre- vs. Post-compensation (Mean ± SD, n=3)



(b)Measurement locations (red boxes) selected along high-curvature regions.

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Figure 22 Statistical measurement results and sampling locations

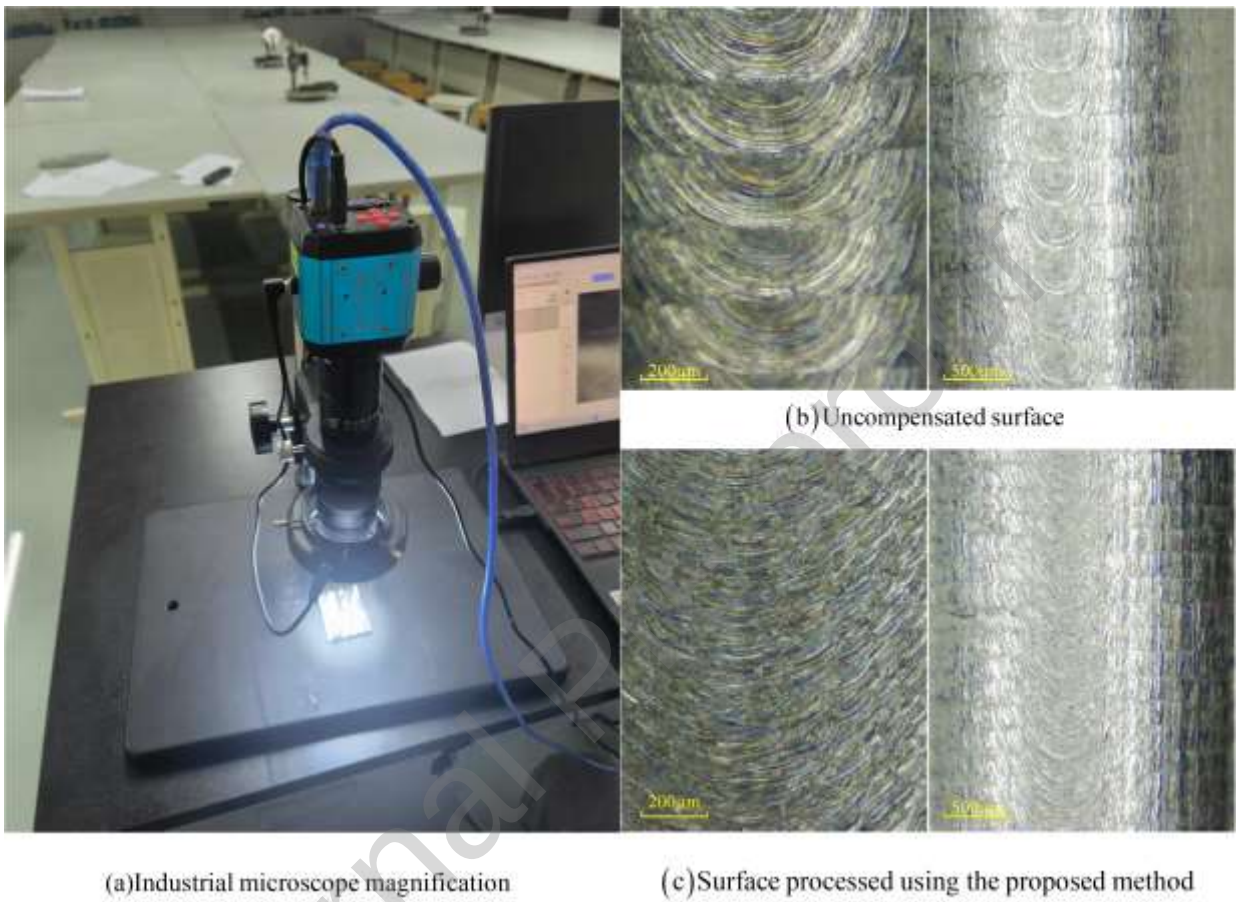

Figure 23 Surface inspection using industrial microscope magnification

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Specifically, regarding machining accuracy, the maximum profile error decreased from 42.310 μm ± 1.687 μm to 3.520 μm ± 0.316 μm, while the average profile error dropped significantly from 14.320 μm ± 0.856 μm to 1.956 μm ± 0.233 μm. In terms of surface quality, the maximum roughness improved from 6.094 μm ± 0.302 μm to 2.273 μm ± 0.157 μm, and the average roughness improved from 1.120 μm ± 0.090 μm to 0.510 μm ± 0.056 μm. To ensure these measurements are representative of the most challenging machining conditions, the sampling locations were strategically selected along the high-curvature regions of the workpiece (crests and troughs), as marked by the red boxes in Figure 22(b). These results collectively confirm that the optimized interpolation strategy significantly enhances both machining precision and surface integrity.

The setup for industrial microscope magnification used to inspect the surface quality of the workpieces is shown in Figure 23(a). To inspect surface details at different scales, Figures 23(b) and (c) present the machining results at two magnification levels: a high-magnification view (scale bar: 200 μm) for observing micro-textures, and a low-magnification view (scale bar: 500 μm) for assessing global uniformity. The uncompensated surface in Figure 23(b) exhibits rough tool marks and an uneven texture. In contrast, the surface processed by the proposed method (Figure 23(c)) appears significantly smoother and

more uniform at both scales, verifying the substantial improvement in surface quality.

6 Discussion

The core contribution of this paper is the proposal and validation of a vector error coupling model, which fundamentally refines the decision-making logic for adaptive interpolation in five-axis machining. While traditional adaptive strategies typically use local curvature as the sole proxy for error risk, the analysis and experimental results of this study demonstrate that high curvature does not always correlate with high geometric error. The key finding is that the final machining accuracy is contingent upon the complex vectorial interaction between the TCP chord height error and the TAV swing error. As revealed by the model, this interaction is critically linked to the local surface geometry's concavity or convexity: errors tend to be "additive" in concave regions and "subtractive" in convex regions.

The physical machining experiments (Figures 19, 20) provide compelling support for the "dual-check" strategy. When processing 6061 aluminum alloy, our method achieved a 91.7% reduction in maximum measured error (from 42.31 to 3.52 μm) and a 54% improvement in average surface roughness (Ra), successfully eliminating the visible tool marks seen in traditional methods (Figure 23). This real-world validation of improved surface integrity is directly corroborated by our simulation results (Figure 15), which show that a "Single Curvature Method" fails in additive coupling regions, whereas our proposed "Hybrid (Dual-Check)" strategy successfully maintains the error within tolerance under all conditions.

Furthermore, the superiority of this strategy lies not only in its robustness but also in its efficiency. By safely permitting the use of high-efficiency LERP in subtractive coupling regions (even when curvature is high), the method achieves a reduction in total computation time of nearly 40% without compromising precision. This validates the significant advantage of analyzing error coupling over analyzing curvature alone. It is also worth noting the efficiency advantage over the 'Single Curvature Method.' To strictly guarantee accuracy, a traditional curvature-based strategy must conservatively switch to the computationally expensive NURBS mode in all high-curvature regions. In contrast, the proposed dual-check strategy intelligently identifies 'safe' subtractive coupling zones (convex surfaces) within these high-curvature regions and retains the efficient LERP mode. This selective optimization allows the proposed method to avoid unnecessary high-precision interpolation, thereby achieving higher overall efficiency than the Single Curvature Method. From an engineering implementation perspective, the computational overhead introduced by the dual-check logic is negligible. The geometric checks (calculating $\cos\theta$ via vector dot products) involve only basic algebraic operations with O(1) complexity, consuming microseconds. This 'decision cost' is far outweighed by the substantial 'execution savings' achieved by avoiding computationally intensive NURBS interpolation in subtractive coupling regions. Consequently, the proposed algorithm is completely feasible for real-time implementation on modern CNC controllers.

Additionally, the proposed dual-check strategy demonstrates excellent robustness against varying manufacturing requirements. Since the error tolerance serves as an adjustable input parameter in the decision logic (Sub-Decision 1), the system can seamlessly adapt to different precision standards. This capability is quantitatively verified in Figure 24, which presents the statistical results obtained from the simulation of the entire freeform surface. As illustrated by the orange bars, the maximum total error strictly adheres to the preset limit across the entire range (1-10 μm). Specifically, under stricter tolerances (e.g., 1 μm), the algorithm effectively tightens the switching criteria, triggering the high-precision NURBS+SLERP mode more frequently to ensure rigorous adherence to profile accuracy. Conversely, as the tolerance relaxes (e.g., 10 μm), the average computation time (blue curve) decreases monotonically, confirming that the system automatically expands the utilization of the efficient LERP mode to maximize material removal rates without violating the specified constraints.

Beyond adaptability to user settings, the robustness of the proposed vector error-coupling model was further evaluated against typical input disturbances, specifically regarding waypoint discretization, geometric noise, and small-feature curvature estimation. First, regarding waypoint discretization, the proposed strategy exhibits a self-correcting mechanism. If the waypoint density is insufficient (sparse discretization), the calculated local chord height error (δ) significantly increases. This automatically triggers the first decision threshold (κ > threshold), forcing the system to switch to the high-precision NURBS mode, which mathematically reconstructs the ideal curve to compensate for the lack of discrete data points. Second, concerning geometric noise (e.g., measurement jitter), the dual-check logic relies on the vectorial relationship ($\cos\theta$) rather than scalar magnitude alone. Random noise typically generates high-frequency fluctuations in the error vector direction.

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However, the system only permits the efficient LERP mode if specific 'subtractive coupling' conditions are met AND the total error is within tolerance.

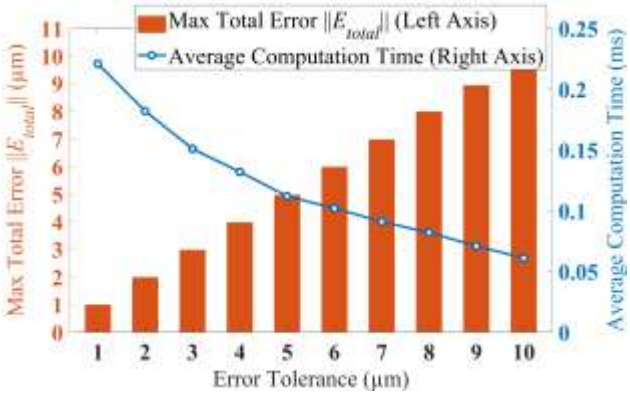

Figure 24 Sensitivity analysis of accuracy and computation time under varying tolerances

Noise-induced fluctuations would likely violate the total error tolerance ($\| E_{\text{total}} \| \leq E_{\text{tolerance}}$), safely reverting the system to the robust NURBS+SLERP mode. Finally, for small-feature curvature estimation, such features inherently possess high local curvature. The proposed curvature-driven filter acts as a sensitive 'first line of defense,' ensuring that these micro-features are immediately identified. Consequently, the system applies smooth NURBS interpolation to preserve feature fidelity, avoiding the kinematic discontinuities often associated with linear interpolation in such scenarios.

While the proposed model performs well, limitations exist. The determination of the coupling mode and its correlation with concavity/convexity relies on local three-point geometric analysis. For highly complex topological structures, such as saddle surfaces or S-shaped flow channels, this simplified local definition may face ambiguity due to direction-dependent curvature variations. Future work could explore more computationally efficient and robust real-time surface topography identification algorithms or incorporate global topological analysis to enhance the model's applicability to more complex geometries. Furthermore, the swing error model (Eq. (14)) currently relies on a fixed tool length (L). In practical long-term machining, tool wear inevitably alters this effective length, potentially affecting the precision of the $\boldsymbol{E}_{\text{swing}}$ prediction. To address this, future industrial implementations should integrate on-machine probing or tool presetting data to dynamically update the value of L in the algorithm, ensuring the model adapts to the tool's physical state. Additionally, the predictive accuracy of the model depends on the precise estimation of the chord height and swing error magnitudes; further refinement of these base error models remains a promising direction for future research.

7 Conclusions

This study proposes and validates a novel "dual-check" adaptive hybrid interpolation strategy based on a vector error coupling model. By fundamentally analyzing the interaction between TCP and TAV errors, the research overcomes the rigid limitations of traditional single-curvature-threshold methods. The specific conclusions are as follows:

(1) A vector error coupling model was established to reveal the intrinsic link between error coupling modes and local surface geometry. This model enables the systematic identification of "additive" and "subtractive" coupling zones, providing a theoretical basis for predicting the total geometric deviation based on surface concavity and convexity.

(2) A refined "dual-check" decision logic was constructed to optimize the selection of interpolation modes. By using curvature as a primary filter and error coupling as a secondary risk assessment, the strategy intelligently maintains high-efficiency linear interpolation in safe subtractive zones or regions where the error is within limits. The system switches to high-precision NURBS and SLERP modes whenever the predicted total error exceeds the allowable tolerance, a condition typically triggered by additive error amplification.

(3) Physical machining experiments and simulations confirmed that the strategy effectively suppresses profile errors and improves surface uniformity. The results demonstrate a superior balance between machining accuracy and computational efficiency, proving that the total geometric error can be strictly maintained within user-defined tolerances under complex

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freeform machining conditions.

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CRediT authorship contribution statement

Haohao Xu: Writing - original draft, Formal analysis. Yueri Cai: Conceptualization, Supervision, Project administration, Writing - review & editing. Shusheng Bi: Investigation, Resources. All authors read and approved the final manuscript.

Declaration of Competing Interest

The authors declare no competing financial interests.

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